UNIVERSIDADE VILA VELHA ESPÍRITO SANTO

DATA SCIENCE PROJECT

WORKFLOW PARTE 1 DO PROJETO

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Link do Colab: Aqui

Link do GitHub: Aqui

Link da Apresentação no Youtube: Aqui





Cenário Fictício:

Prezados(as) alunos(as),

É com entusiasmo que apresento a proposta do **Projeto de Conclusão de Curso (TCC)** para os estudantes do curso de Ciência de Dados da Universidade Vila Velha: o UWine – Clube de Vinhos da UVV.

A proposta do projeto consiste na análise de um Clube de Vinhos universitário fictício, com base em dados simulados, que será totalmente orientado por técnicas e práticas de Ciência de Dados.

Sendo assim, você foi designado como **Consultor em Ciência de Dados** (***Data Science Consultant***) para fazer uma análise completa dos dados dos clientes a partir de um conjunto de dados (Amostra) com mais de 1 milhão de notas fiscais (~ 350 Megabytes). Além disso, há uma entrada diária de aproximadamente umas 5000 novas notas fiscais no sistema.

O objetivo é permitir que cada consultor(a) percorra foco analíticos diferentes dentro do mesmo ecossistema de dados, aplicando modelos de machine learning supervisionado ou não supervisionado, análise preditiva, dashboards interativos, entre outros.

Para garantir organização, profundidade técnica e coerência metodológica no desenvolvimento do ***Projeto UWine***, adotaremos um workflow composto por cinco etapas principais, alinhadas às boas práticas da Ciência de Dados. Cada etapa representa um marco do processo analítico, permitindo que cada analista avance de forma estruturada do entendimento inicial dos dados até a entrega final dos insights.

BIBLIOTECAS E FUNÇÕES E CÓDIGOS

In [ ]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [ ]:
```
# carregando a table45
table45 = pd.read_csv('table45.csv', index_col=0)
```

In [ ]:
```
table45.head()
```

Out[ ]:

	NOTA FISCAL	REGIÃO	SEXO	ESTADO CIVIL	DEPENDENTES	RENDA BRUTO (R$)	OPINIÃO DO CLIENTE
0	7894261	SUDESTE	MASCULINO	CASADO	0	8017.69	Entrega foi boa e rápida na UVVine
1	1250786	SUL	MASCULINO	SOLTEIRO	2	7736.06	Funcionou bem e me atendeu no prazo certo a UV...
2	3639907	SUL	MASCULINO	CASADO	2	3196.50	Serviço eficiente, recomendo o clube: UVVine
3	7210729	SUDESTE	MASCULINO	CASADO	0	3283.15	Foi indiferente minha experiência com a UVVine
4	8141835	SUDESTE	MASCULINO	DESQUITADO(A) / VIÚVO(A)	2	3234.05	Serviço eficiente, recomendo o clube: UVVine

5 rows × 23 columns

In [ ]:
```
table45.shape
```

Out[ ]:
```
(1120000, 23)
```

In [ ]:
```
def optimal_stratified_sample_size(
    df,
    value_column,
    strata_column,
    n_min=100,
    n_max=50000,
    step=100,
    stabilization_threshold=0.01,
    consecutive_points=3,
    plot=True
):
    """
    Estimate the optimal sample size for an infinite stratified population
    using Neyman allocation and consecutive marginal stabilization of the
    stratified standard error.

    Parameters
    ----------
```

```
    df : pandas.DataFrame
        Input dataset.

    value_column : str
        Numerical target variable used to estimate the mean.

    strata_column : str
        Categorical stratification variable.

    n_min : int, default=100
        Minimum total sample size evaluated.

    n_max : int, default=50000
        Maximum total sample size evaluated.

    step : int, default=100
        Increment between consecutive sample sizes.

    stabilization_threshold : float, default=0.01
        Maximum relative reduction accepted for standard error stabilization.
        Example: 0.01 means 1%.

    consecutive_points : int, default=3
        Number of consecutive marginal reductions that must remain below
        the stabilization threshold.

    plot : bool, default=True
        If True, plot the standard error stabilization curve.

    Returns
    -------
    summary : pandas.DataFrame
        Sample allocation per stratum.

    results : dict
        Dictionary containing:
        - optimal_n
        - optimal_standard_error
        - sample_sizes
        - standard_error_curve
        - marginal_reduction
        - stabilization_threshold
        - consecutive_points
        - strata_statistics
    """

    # Keep only the required columns and remove missing values
    data = df[[value_column, strata_column]].dropna().copy()

    # Convert the target variable to numeric
    data[value_column] = pd.to_numeric(data[value_column], errors="coerce")
    data = data.dropna()

    if data.empty:
        raise ValueError("The dataset is empty after removing missing or invalid

    # Estimate stratum weights
    weights = (
        data[strata_column]
        .value_counts(normalize=True)
```

```
    .sort_index()
)

# Estimate stratum standard deviations
stds = (
    data.groupby(strata_column)[value_column]
    .std()
    .sort_index()
)

# Align both series
stds = stds.loc[weights.index]

strata = weights.index.values
Wh = weights.values
Sh = stds.values

if np.any(np.isnan(Sh)):
    raise ValueError(
        "At least one stratum has insufficient observations "
        "to calculate standard deviation."
    )

if np.any(Sh == 0):
    print(
        "Warning: At least one stratum has zero standard deviation. "
        "Neyman allocation may assign a very small sample to this stratum."
    )

# Neyman allocation weights
neyman_weights = Wh * Sh

if neyman_weights.sum() == 0:
    raise ValueError(
        "Neyman weights are zero. Check the variability of the target variab
    )

sample_sizes = np.arange(n_min, n_max + step, step)

standard_errors = []
allocations = []

for n in sample_sizes:

    nh = n * neyman_weights / neyman_weights.sum()

    se = np.sqrt(
        np.sum(
            (Wh ** 2) * (Sh ** 2) / nh
        )
    )

    standard_errors.append(se)
    allocations.append(nh)

standard_errors = np.array(standard_errors)
allocations = np.array(allocations)

# Marginal relative reduction of the standard error
marginal_reduction = (
```

```
        np.abs(np.diff(standard_errors)) / standard_errors[:-1]
    )

    stable_sequence_index = None

    for i in range(0, len(marginal_reduction) - consecutive_points + 1):

        window = marginal_reduction[i:i + consecutive_points]

        if np.all(window < stabilization_threshold):
            stable_sequence_index = i
            break

    if stable_sequence_index is None:
        raise ValueError(
            "No consecutive stabilization point found within the evaluated inter
            "Try increasing n_max, increasing step, or using a larger threshold.
        )

    optimal_index = stable_sequence_index + consecutive_points

    optimal_n = sample_sizes[optimal_index]
    optimal_se = standard_errors[optimal_index]
    optimal_allocation = allocations[optimal_index]

    summary = pd.DataFrame({
        "Stratum": strata,
        "Population Weight (Wh)": Wh,
        "Std. Deviation (Sh)": Sh,
        "Neyman Weight (Wh x Sh)": neyman_weights,
        "Allocated Sample (nh)": np.round(optimal_allocation).astype(int)
    })

    summary["Allocated Sample (%)"] = (
        summary["Allocated Sample (nh)"] /
        summary["Allocated Sample (nh)"].sum()
    )

    strata_statistics = pd.DataFrame({
        "Stratum": strata,
        "Wh": Wh,
        "Sh": Sh,
        "Wh_x_Sh": neyman_weights
    })

    if plot:

        plt.figure(figsize=(11, 6))

        plt.plot(
            sample_sizes,
            standard_errors,
            linewidth=3,
            label="Stratified standard error"
        )

        plt.scatter(
            optimal_n,
            optimal_se,
            s=120,
```

```
            label="Optimal sample size"
        )

        plt.axvline(
            optimal_n,
            linestyle="--"
        )

        plt.text(
            optimal_n,
            optimal_se,
            f"  n = {optimal_n}\n  SE = {optimal_se:.4f}",
            verticalalignment="bottom"
        )

        plt.xlabel("Total Sample Size (n)")
        plt.ylabel("Stratified Standard Error")
        plt.title("Consecutive Marginal Stabilization of the Standard Error")
        plt.grid(True)
        plt.legend()
        plt.show()

    results = {
        "optimal_n": int(optimal_n),
        "optimal_standard_error": float(optimal_se),
        "sample_sizes": sample_sizes,
        "standard_error_curve": standard_errors,
        "marginal_reduction": marginal_reduction,
        "stabilization_threshold": stabilization_threshold,
        "consecutive_points": consecutive_points,
        "strata_statistics": strata_statistics
    }

    return summary, results
```

In [ ]:

```
def validar_tamanho_amostra(dados, n, n_simulacoes=1000, z=1.96, seed=42):
    """
    Prova que o tamanho de amostra `n` estima a média real do grupo sem
    distorção, testando 3 propriedades independentes:
      1) Viés: a média das amostras repetidas deve convergir para a
         média real da população (prova ausência de distorção).
      2) Precisão: margem de erro do IC, em % da média.
      3) Cobertura: % de vezes que o IC de 95%, construído a partir de
         uma amostra, realmente contém a média real, em repetições
         independentes.
    """
    rng = np.random.default_rng(seed)
    media_real = dados.mean()

    medias_amostras = []
    cobriu = 0
    for _ in range(n_simulacoes):
        amostra = rng.choice(dados, size=n, replace=False)
        media_amostra = amostra.mean()
        se = amostra.std(ddof=1) / np.sqrt(n)
        ic_inf, ic_sup = media_amostra - z * se, media_amostra + z * se
        medias_amostras.append(media_amostra)
        if ic_inf <= media_real <= ic_sup:
            cobriu += 1
```

```
    medias_amostras = np.array(medias_amostras)
    vies_pct = abs(medias_amostras.mean() - media_real) / media_real * 100
    margem_pct = z * (dados.std(ddof=1) / np.sqrt(n)) / media_real * 100
    cobertura_pct = cobriu / n_simulacoes * 100

    return {
        "media_real": media_real,
        "media_estimada": medias_amostras.mean(),
        "vies_%": vies_pct,
        "margem_erro_%": margem_pct,
        "cobertura_%": cobertura_pct,
    }
```

In [ ]:

```
def AtividadeA1_1():
    """
    Atividade A1.1 - Tamanho ideal da amostra.

    Passo 1: Procedimento empírico (optimal_sample_size, do professor),
             rodado para ESSENTIAL, VIP, PRIME e GERAL -> 4 gráficos.
    Passo 2: Procedimento teórico-empírico (optimal_stratified_sample_size,
             do professor), com os 3 tipos de conta como estratos
             (alocação de Neyman).
    Passo 3: Tamanho ideal ADOTADO, por busca via bootstrap real: testamos
             tamanhos candidatos, medimos a margem do IC 95% por
             reamostragem em cada um, e adotamos o maior "menor n" entre
             os 4 grupos (cenário mais exigente).
    Passo 4: Validação final (viés, precisão e cobertura real) no n
             adotado, respondendo definitivamente: Qual é o tamanho
             ideal da amostra, e ele distorce a média de algum grupo?
    """
    target = 'TOTAL (R$)'

    grupos = {
        'ESSENTIAL': table45[table45['TIPO DA CONTA'] == 'ESSENTIAL'],
        'VIP':       table45[table45['TIPO DA CONTA'] == 'VIP'],
        'PRIME':     table45[table45['TIPO DA CONTA'] == 'PRIME'],
        'GERAL':     table45,
    }

    # --- Passo 1: Procedimento empírico (4 gráficos) ---
    print("="*70)
    print("PASSO 1: Procedimento empírico (optimal_sample_size)")
    print("="*70)
    for nome, sub in grupos.items():
        resultado = optimal_sample_size(sub, target)
        print(f"{nome}: n ideal = {resultado['optimal_sample_size']}, "
              f"erro-padrão = R$ {resultado['standard_error']:.4f}")

    # --- Passo 2: Procedimento teórico-empírico (Neyman) ---
    print("\n" + "="*70)
    print("PASSO 2: Procedimento teórico-empírico (optimal_stratified_sample_siz
    print("="*70)
    summary, resultado_estrat = optimal_stratified_sample_size(
        table45, target, 'TIPO DA CONTA'
    )
    print(summary)
    print(f"\nn ideal (estratificado) = {resultado_estrat['optimal_n']}")

    # --- Passo 3: Tamanho ideal adotado, por busca via bootstrap real ---
    print("\n" + "="*70)
```

```
    print("PASSO 3: Busca do tamanho ideal por bootstrap (margem de IC <= 1%)")
    print("="*70)
    rng = np.random.default_rng(42)
    candidatos = [1000, 2500, 5000, 7500, 10000, 15000, 20000, 25000, 30000, 400
    margem_alvo = 0.01

    n_por_grupo = {}
    for nome, sub in grupos.items():
        dados = pd.to_numeric(sub[target], errors='coerce').dropna().to_numpy()
        n_escolhido = None
        for n in candidatos:
            idx = rng.integers(0, dados.shape[0], size=(300, n))
            medias = dados[idx].mean(axis=1)
            ic_inf, ic_sup = np.percentile(medias, [2.5, 97.5])
            margem_pct = (ic_sup - ic_inf) / 2 / medias.mean()
            if margem_pct <= margem_alvo:
                n_escolhido = n
                break
        n_por_grupo[nome] = n_escolhido
        print(f"{nome}: menor n com margem <= {margem_alvo:.0%} = {n_escolhido:,

    n_final = max(n_por_grupo.values())
    print(f"\nTamanho ideal da amostra adotado (maior exigência entre os grupos)

    # --- Passo 4: Validação final (viés, precisão, cobertura) ---
    print("\n" + "="*70)
    print(f"PASSO 4: Validação final no n adotado (n = {n_final:,})")
    print("="*70)
    validacoes = {}
    for nome, sub in grupos.items():
        dados = pd.to_numeric(sub[target], errors='coerce').dropna().to_numpy()
        v = validar_tamanho_amostra(dados, n_final)
        validacoes[nome] = v
        print(f"{nome:10s}: média real = R$ {v['media_real']:>9,.2f} | "
              f"média estimada = R$ {v['media_estimada']:>9,.2f} | "
              f"viés = {v['vies_%']:.3f}% | "
              f"margem = ±{v['margem_erro_%']:.2f}% | "
              f"cobertura = {v['cobertura_%']:.1f}%")

    print(f"\nRESPOSTA FINAL: o tamanho ideal da amostra é {n_final:,}, "
          f"validado como não-viesado, preciso e estatisticamente confiável "
          f"em todos os grupos analisados.")

    return {
        "estratificado": (summary, resultado_estrat),
        "n_por_grupo": n_por_grupo,
        "n_final": n_final,
        "validacoes": validacoes,
    }
```

In [ ]:

```
def AtividadeA1_2():
    """
    Atividade A1.2 - Análise de Outliers.

    Método: IQR (intervalo interquartil), aplicado separadamente por
    TIPO DA CONTA (ESSENTIAL, VIP, PRIME) - já vimos que, sem separar,
    gastos altos legítimos de PRIME seriam confundidos com outlier só
    por serem comparados à base inteira (dominada por ESSENTIAL).

    Decisão adotada: MANTER os outliers. São gastos legítimos de
```

```
    clientes de alto consumo, não erros de digitação ou falha de
    sistema - removê-los distorceria o comportamento real de consumo,
    especialmente no grupo PRIME.
    """
    target = 'TOTAL (R$)'
    grupos = ['ESSENTIAL', 'VIP', 'PRIME']

    resumo = []
    fig, eixos = plt.subplots(1, len(grupos), figsize=(14, 5))

    for i, tipo in enumerate(grupos):
        dados = table45[table45['TIPO DA CONTA'] == tipo][target]
        q1, q3 = dados.quantile(0.25), dados.quantile(0.75)
        iqr = q3 - q1
        limite_inf = max(q1 - 1.5 * iqr, 0)  # gasto não pode ser negativo
        limite_sup = q3 + 1.5 * iqr
        outliers = dados[(dados < limite_inf) | (dados > limite_sup)]

        resumo.append({
            'Tipo': tipo,
            'Q1': round(q1, 2),
            'Q3': round(q3, 2),
            'IQR': round(iqr, 2),
            'Limite inferior': round(limite_inf, 2),
            'Limite superior': round(limite_sup, 2),
            'Qtd outliers': len(outliers),
            '% outliers': round(len(outliers) / len(dados) * 100, 2),
        })

        eixos[i].boxplot(dados, vert=True)
        eixos[i].set_title(tipo)
        eixos[i].set_ylabel('TOTAL (R$)')

    plt.suptitle('Análise de Outliers - TOTAL (R$) por Tipo de Conta')
    plt.tight_layout()
    plt.show()

    resumo_df = pd.DataFrame(resumo)
    print(resumo_df.to_string(index=False))

    print("\nDECISÃO: Outliers MANTIDOS na base.")
    print("Justificativa: representam gastos legítimos de clientes de alto")
    print("consumo, não erros de coleta ou digitação. Removê-los distorceria")
    print("o comportamento real de consumo de cada tipo de conta.")

    return resumo_df
```

In [ ]:

```
# Sorteio da Amostra com o Tamanho Ideal (n = 40.000)

n_ideal = 40000

sample_1 = table45[table45['TIPO DA CONTA'] == 'ESSENTIAL'].sample(
    n=n_ideal, replace=False, random_state=42
)['TOTAL (R$)']

sample_2 = table45[table45['TIPO DA CONTA'] == 'PRIME'].sample(
    n=n_ideal, replace=False, random_state=42
)['TOTAL (R$)']
```

```
print(f"SAMPLE 1 (ESSENTIAL): {len(sample_1):,} linhas, média = R$ {sample_1.mea
print(f"SAMPLE 2 (PRIME): {len(sample_2):,} linhas, média = R$ {sample_2.mean():
```

```
SAMPLE 1 (ESSENTIAL): 40,000 linhas, média = R$ 416.21
SAMPLE 2 (PRIME): 40,000 linhas, média = R$ 2,154.59
```

In [ ]:

```
from scipy.stats import anderson

def Normality_Test(data, nome):
    """
    Teste de Normalidade de Anderson-Darling,
    checando vários níveis de confiança de uma vez.
    """
    result = anderson(data)
    print(f"{nome} - DISTRIBUIÇÃO AMOSTRAL: stat = {result.statistic:.3f}")
    for i in range(len(result.critical_values)):
        sl, cv = result.significance_level[i], result.critical_values[i]
        if result.statistic < cv:
            print(f"  Intervalo de Confiança ({sl:.1f}%): Provavelmente Gaussian
        else:
            print(f"  Intervalo de Confiança ({sl:.1f}%): Provavelmente NÃO Gaus
    print()


def AtividadeA1_3_1():
    """
    Atividade A1.3.1 - Teste de Normalidade (Anderson-Darling).
    SAMPLE 1 = ESSENTIAL, SAMPLE 2 = PRIME, ambas com n = 40.000
    (tamanho ideal definido na A1.1).
    """
    Normality_Test(sample_1, "SAMPLE 1 (ESSENTIAL)")
    Normality_Test(sample_2, "SAMPLE 2 (PRIME)")
```

In [ ]:

```
# PLOT DO GRÁFICO DE BOOTSTRAP (A1.3.2)
from scipy.stats import bootstrap

def plot_bootstrap_correlacao(sample_1, sample_2, n_resamples=1000, seed=42):
    """
    Gráfico Bootstrap para o Teste de Independência (A1.3.2):
    distribuição bootstrap da correlação de Spearman entre as duas
    amostras, com o Intervalo de Confiança de 95% marcado.
    """
    rng = np.random.default_rng(seed)

    def spearman_stat(x, y):
        return spearmanr(x, y).statistic

    resultado = bootstrap(
        (sample_1.to_numpy(), sample_2.to_numpy()), spearman_stat,
        paired=True, vectorized=False, confidence_level=0.95,
        random_state=rng, n_resamples=n_resamples, method='percentile'
    )
    dist = resultado.bootstrap_distribution

    plt.figure(figsize=(9, 5))
    plt.hist(dist, bins=30, density=True, alpha=0.7, color='steelblue')
    plt.axvline(0, color='black', linestyle='-', linewidth=1.5, label='Zero (sem
    plt.axvline(resultado.confidence_interval.low, color='crimson', linestyle='-
    plt.axvline(resultado.confidence_interval.high, color='crimson', linestyle='
    plt.title('Bootstrap da Correlação de Spearman entre SAMPLE 1 e SAMPLE 2')
```

```
    plt.xlabel('Correlação de Spearman (reamostras)')
    plt.ylabel('Densidade')
    plt.legend()
    plt.show()

    return resultado
```

In [ ]:
```
from scipy.stats import pearsonr, spearmanr, kendalltau

def AtividadeA1_3_2():
    """
    Atividade A1.3.2 - Teste de Independência (Correlação + Gráfico Bootstrap).
    SAMPLE 1 = ESSENTIAL, SAMPLE 2 = PRIME, n = 40.000 cada.

    Nota: o Chi-quadrado exige dados categóricos em tabela de
    contingência, não valores contínuos brutos - por isso não é
    aplicado aqui (produziria resultado degenerado).
    """
    resultados = {
        'Pearson':  pearsonr(sample_1, sample_2),
        'Spearman': spearmanr(sample_1, sample_2),
        'Kendall':  kendalltau(sample_1, sample_2),
    }

    for nome, (stat, p) in resultados.items():
        print(f"{nome}: stat = {stat:.4f}, p-valor = {p:.4f}")

    # Resultado + Plot
    resultado_bootstrap = plot_bootstrap_correlacao(sample_1, sample_2)
    print(f"\nIC 95% (bootstrap) da correlação de Spearman: {resultado_bootstrap

    return resultados, resultado_bootstrap
```

In [ ]:
```
import seaborn as sns
from scipy.stats import bootstrap

def bootstrap_distribution_eda(data, metric=np.mean, bins=30, confidence_level=0
    """
    Bootstrap da distribuição amostral de uma estatística (função do
    professor, adaptada com 2 ajustes para funcionar com n=40.000):
      - method='percentile' em vez do padrão (BCa trava por memória
        com amostras desse tamanho)
      - n_resamples=1000 em vez do padrão 9.999 (mais leve, IC já
        fica estável)
    """
    rng = np.random.default_rng(seed)
    res = bootstrap((data,), metric, confidence_level=confidence_level,
                    random_state=rng, method='percentile', n_resamples=n_resamp
    dist = res.bootstrap_distribution

    plt.figure(figsize=(10, 6))
    sns.histplot(dist, bins=bins, kde=True, stat="density")
    plt.axvline(np.mean(dist), linestyle='-', label='Bootstrap Mean')
    plt.axvline(res.confidence_interval.low, linestyle='--', label='CI Lower')
    plt.axvline(res.confidence_interval.high, linestyle='--', label='CI Upper')
    plt.title("Bootstrap Sampling Distribution")
    plt.xlabel("Statistic Value")
    plt.ylabel("Density")
    plt.legend()
```

```
    plt.show()

    print("CONFIDENCE INTERVAL")
    print(f"LOW  : {res.confidence_interval.low:.4f}")
    print(f"HIGH : {res.confidence_interval.high:.4f}")
    print(f"SE   : {res.standard_error:.4f}")
    return res
```

In [ ]:
```
def AtividadeA1_3_3():
    """
    Atividade A1.3.3 - Teste de Distribuição (+ Gráfico Bootstrap).
    SAMPLE 1 = ESSENTIAL, SAMPLE 2 = PRIME, n = 40.000 cada.
    """
    resultados = {
        'ttest_ind (paramétrico)':  ttest_ind(sample_1, sample_2),
        'f_oneway (paramétrico)':   f_oneway(sample_1, sample_2),
        'mannwhitneyu (não-paramétrico)': mannwhitneyu(sample_1, sample_2),
        'kruskal (não-paramétrico)':      kruskal(sample_1, sample_2),
    }

    for nome, resultado in resultados.items():
        print(f"{nome}: stat = {resultado.statistic:.4f}, p-valor = {resultado.p

    print("\n--- Gráfico Bootstrap: SAMPLE 1 (ESSENTIAL) ---")
    res_1 = bootstrap_distribution_eda(sample_1.to_numpy(), confidence_level=0.9

    print("\n--- Gráfico Bootstrap: SAMPLE 2 (PRIME) ---")
    res_2 = bootstrap_distribution_eda(sample_2.to_numpy(), confidence_level=0.9

    return resultados, (res_1, res_2)
```

In [ ]:
```
def AtividadeA1_3_4():
    """
    Atividade A1.3.4 - Teste das Médias Amostrais (+ Gráfico Bootstrap).
    SAMPLE 1 = ESSENTIAL, SAMPLE 2 = PRIME, n = 40.000 cada.

    Foco específico: as médias das duas amostras são estatisticamente
    iguais? (ttest_ind já foi calculado na A1.3.3, mas é o teste
    central desta atividade - reaproveitado aqui.)
    """
    resultado = ttest_ind(sample_1, sample_2)
    print(f"ttest_ind: stat = {resultado.statistic:.4f}, p-valor = {resultado.pv
    print(f"Média SAMPLE 1 (ESSENTIAL): R$ {sample_1.mean():,.2f}")
    print(f"Média SAMPLE 2 (PRIME): R$ {sample_2.mean():,.2f}")

    print("\n--- Gráfico Bootstrap: SAMPLE 1 (ESSENTIAL) ---")
    res_1 = bootstrap_distribution_eda(sample_1.to_numpy(), confidence_level=0.9

    print("\n--- Gráfico Bootstrap: SAMPLE 2 (PRIME) ---")
    res_2 = bootstrap_distribution_eda(sample_2.to_numpy(), confidence_level=0.9

    return resultado, (res_1, res_2)
```

In [ ]:
```
# Função importada do Topico I

def statistic(_dataset):
    """
      Compute descriptive statistics for a numerical dataset.
```

```
    This function extends the standard pandas `describe()` method
    by removing the count metric and adding:
    - Range (max - min)
    - Mode

    The output is a table of descriptive statistics commonly used
    in Exploratory Data Analysis (EDA).

    Parameters
    ----------
    _dataset : pandas.Series or pandas.DataFrame
        Input dataset containing numerical values.

    Returns
    -------
    pandas.DataFrame
        A DataFrame with descriptive statistics (excluding count),
        including range and mode, sorted and rounded to two decimals.
    """

    # Generate basic descriptive statistics using pandas
    # (count, mean, std, min, quartiles, and max)
    _describe = _dataset.describe()

    # Remove the 'count' row, as it is not used in this analysis
    _describe = _describe.drop('count')

    # Compute the range for each numerical column
    # Range = maximum value minus minimum value
    _describe.loc['range'] = _describe.loc['max'] - _describe.loc['min']

    # Compute the mode for each numerical column
    # stats.mode returns both mode and frequency;
    # [0] extracts only the modal value
    _describe.loc['mode'] = stats.mode(
        _dataset,
        keepdims=False,
        nan_policy='omit'
    )[0]

    # Sort the statistics (index) and round all values to two decimal places
    _describe = np.around(_describe.sort_values(), 2)

    # Return the final descriptive statistics table
    return _describe
```

In [ ]:

```
# Função importada do Topico I

def PlotarStatistic(dataset, yLabel=None):
    """
      Plot descriptive statistics of a dataset and highlight
      measures of central tendency (mean, median, and mode).

      This function computes descriptive statistics using a
      previously defined `statistic()` function, generates
      a plot with summary measures, and prints the numerical
      descriptive table.

      Parameters
      ----------
```

```
    dataset : pandas.Series or pandas.DataFrame
        Input dataset containing numerical values.
        It is expected that the statistic() function
        can process this object.

    yLabel : str, optional
        Label for the y-axis of the plot.

    Returns
    -------
    None
        Displays a plot and prints descriptive statistics.
    """

    # Compute descriptive statistics using a custom function
    _describe = statistic(dataset)

    # Símbolos da estatística estimada da população
    mean_mu = r'MÉDIA:  $\overline{\mu}$ = '
    mean_mu += f'{_describe["mean"]: .2f}'

    median_mu = r'MEDIANA: $\tilde{\mu}$ = '
    median_mu += f'{_describe["50%"]: .2f}'

    mode_mu = r'MODA: $\hat{\mu}$ = '
    mode_mu += f'{_describe["mode"]: .2f}'

    enter = '\n'

    label = mean_mu + enter + median_mu + enter + mode_mu

    # Apply the statistic function as an aggregation and plot the result
    dataset.agg(statistic).plot(
        figsize=[15, 6],
        label='MEDIDAS DE RESUMO: '
    )

    plt.plot(
        list(_describe.index).index('mean'),
        _describe["mean"],
        'D',
        label=mean_mu
    )

    plt.plot(
        list(_describe.index).index('50%'),
        _describe["50%"],
        'D',
        label=median_mu
    )

    plt.plot(
        list(_describe.index).index('mode'),
        _describe["mode"],
        'D',
        label=mode_mu
    )

    plt.title('ESTATÍSTICA DESCRITIVA DO DATASET')
    plt.legend()
```

```
    plt.ylabel(yLabel)
    plt.show()

    print('ESTATÍSTICA DESCRITIVA DO DATASET:')
    print(_describe)
```

In [ ]:
```
from scipy import stats
import matplotlib.pyplot as plt

def AtividadeA2_1():
    """
    Atividade A2.1 - Estatística Descritiva das Variáveis Quantitativas.

    Usa as funções statistic() e PlotarStatistic() do Tópico I (link do
    professor). Aplicadas sobre a base completa (table45), não as
    amostras ESSENTIAL/PRIME da A1.3 - aqui é uma visão geral do
    dataset, não uma comparação entre grupos.

    Variáveis com gráfico individual: RENDA BRUTO, TOTAL (R$) e
    NOTA DE SATISFAÇÃO (as 3 variáveis quantitativas de negócio).
    Perfil químico do vinho (13 colunas MEAN:): tabela consolidada,
    sem gráfico individual (13 gráficos separados seria redundante).
    """
    for col in ['RENDA BRUTO (R$)', 'TOTAL (R$)', 'NOTA DE SATISFAÇÃO (%)']:
        PlotarStatistic(table45[col], yLabel=col)

    wine_cols = [c for c in table45.columns if c.startswith('MEAN:')]
    tabela_vinho = pd.DataFrame({col: statistic(table45[col]) for col in wine_co
    print("PERFIL QUÍMICO DO VINHO (consolidado):")
    print(tabela_vinho.round(2))

    return tabela_vinho
```

In [ ]:
```
def pareto_chart(dataset, titulo=None, figsize=(10, 6)):
    """
    Gráfico de Pareto: barras de frequência (decrescente) + linha de
    percentual acumulado, com referência nos 80%.
    """
    contagem = dataset.value_counts().sort_values(ascending=False)
    pct_acumulado = contagem.cumsum() / contagem.sum() * 100

    fig, ax1 = plt.subplots(figsize=figsize)
    ax1.bar(contagem.index.astype(str), contagem.values, color='steelblue')
    ax1.set_ylabel('Frequência')
    ax1.tick_params(axis='x', rotation=45)

    ax2 = ax1.twinx()
    ax2.plot(contagem.index.astype(str), pct_acumulado.values, color='crimson',
    ax2.axhline(80, color='gray', linestyle='--', linewidth=1)
    ax2.set_ylabel('% Acumulado')
    ax2.set_ylim(0, 110)

    plt.title(titulo or f'Gráfico de Pareto - {dataset.name}')
    plt.tight_layout()
    plt.show()

    return contagem, pct_acumulado
```

```
def AtividadeA2_2():
    """
    Atividade A2.2 - Estatística Descritiva das Variáveis Qualitativas.
    Técnica: Gráfico de Pareto (frequência + % acumulado), para cada
    variável qualitativa: REGIÃO, SEXO, ESTADO CIVIL, TIPO DA CONTA,
    DEPENDENTES e OPINIÃO DO CLIENTE.
    """
    colunas_qualitativas = ['REGIÃO', 'SEXO', 'ESTADO CIVIL', 'TIPO DA CONTA',
                            'DEPENDENTES', 'OPINIÃO DO CLIENTE']

    resultados = {}
    for col in colunas_qualitativas:
        contagem, pct_acumulado = pareto_chart(table45[col])
        resultados[col] = (contagem, pct_acumulado)
        print(f"{col}: {len(contagem)} categorias; "
              f"{(pct_acumulado <= 80).sum() + 1} delas já somam 80% do total.\n

    return resultados
```

In [ ]:

```
def AtividadeA3_1():
    """
    Atividade A3.1 - Discretizar a Variável Qualitativa (OPINIÃO DO CLIENTE).

    Classifica cada uma das 25 frases fixas em 1 das 5 notas de
    satisfação, com base no tom semântico da frase (a coluna numérica
    NOTA DE SATISFAÇÃO (%) não ajuda aqui - é praticamente idêntica em
    todas as 25 frases, sem relação com o conteúdo da opinião).

    NOTA 1: Muito Insatisfeito
    NOTA 2: Insatisfeito
    NOTA 3: Indiferente
    NOTA 4: Satisfeito
    NOTA 5: Muito Satisfeito
    """
    mapa_satisfacao = {
        'Serviço impecável, adorei o clube:  UVVine': 5,
        'Superou expectativas o atendimento na UVVine': 5,
        'Ótimo! Tudo ocorreu maravilhosamente bem com a  UVVine': 5,
        'Produto perfeito, chegou antes do prazo de previsão. Top demais a UVVin
        'Excelente experiência, recomendo muito a UVVine': 5,

        'Entrega foi boa e rápida na UVVine': 4,
        'Funcionou bem e me atendeu no prazo certo a UVVine': 4,
        'Fiquei satisfeito com o serviço da UVVine': 4,
        'Serviço eficiente, recomendo o clube:  UVVine': 4,
        'Gostei do atendimento prestado pela UVVine': 4,

        'Nada digno de nota com a UVVine': 3,
        'Foi indiferente minha experiência com a UVVine': 3,
        'Não tenho muito a dizer sobre a UVVine': 3,
        'Sem opinião formada sobre a UVVine': 3,
        'Compro exponenciamente, opinião neutra sobre UVVine': 3,

        'Atendimento comum, poderia melhorar em cordialidade a UVVine': 2,
        'Nada demais, serviço regular na UVVine': 2,
        'Experiência regular, não atendeu a política de cashback da UVVine': 2,
        'Foi aceitável, esperava mais da UVVine': 2,
        'Razoável, com margem de melhora no preço médio da UVVine': 2,

        'Senti que faltou qualidade no atendimento da UVVine': 1,
```

```
        'Tive uma experiência ruim com a UVVine': 1,
        'Não gostei do atendimento na UVVine': 1,
        'Serviço deixou a desejar na UVVine': 1,
        'Produto chegou atrasado na UVVine': 1,
    }

    table45['NOTA'] = table45['OPINIÃO DO CLIENTE'].map(mapa_satisfacao)

    nao_mapeados = table45['NOTA'].isnull().sum()
    print(f"Valores não mapeados (deveria ser 0): {nao_mapeados}")
    print()

    distribuicao = table45['NOTA'].value_counts().sort_index(ascending=False)
    distribuicao_pct = (distribuicao / distribuicao.sum() * 100).round(2)

    rotulos = {5: 'Muito Satisfeito', 4: 'Satisfeito', 3: 'Indiferente',
               2: 'Insatisfeito', 1: 'Muito Insatisfeito'}

    for nota, qtd in distribuicao.items():
        print(f"NOTA {nota} ({rotulos[nota]}): {qtd:,} ({distribuicao_pct[nota]}

    plt.figure(figsize=(8, 5))
    plt.bar([rotulos[n] for n in distribuicao.index], distribuicao.values, color
    plt.title('Distribuição das Notas de Satisfação')
    plt.ylabel('Frequência')
    plt.xticks(rotation=20)
    plt.tight_layout()
    plt.show()

    return table45['NOTA']
```

In [ ]:
```
def AtividadeA3_2():
    """
    Atividade A3.2 - Análise Bootstrap do Resultado da Pesquisa de Satisfação.

    Calcula Erro-Padrão, Intervalo de Confiança e o Gráfico Bootstrap
    sobre a média da NOTA de satisfação (resultado geral, não separado
    por grupo - isso fica pra A3.3). Usa uma amostra de 40.000 sorteada
    da base inteira (mesma lógica de "amostra ideal" do GERAL na A1).
    """
    sample_geral = table45.sample(n=40000, replace=False, random_state=42)['NOTA

    print(f"Média amostral da NOTA: {sample_geral.mean():.4f}")

    resultado = bootstrap_distribution_eda(sample_geral.to_numpy(), confidence_l

    print(f"\nErro-Padrão (bootstrap): {resultado.standard_error:.4f}")
    print(f"IC 95%: [{resultado.confidence_interval.low:.4f} ; {resultado.confid

    return sample_geral, resultado
```

In [ ]:
```
def AtividadeA3_3():
    """
    Atividade A3.3 - Resultado Final da Pesquisa de Satisfação.
    Percentual de cada NOTA (1 a 5), Geral e Por Sexo.
    """
    geral = table45['NOTA'].value_counts(normalize=True).sort_index(ascending=Fa
    por_sexo = (table45.groupby('SEXO')['NOTA'].value_counts(normalize=True)
                .unstack().sort_index(axis=1, ascending=False) * 100)
```

```
    print("RESULTADO GERAL (%):")
    print(geral.round(2))
    print()
    print("RESULTADO POR SEXO (%):")
    print(por_sexo.round(2))

    rotulos = {5: 'Muito Satisfeito', 4: 'Satisfeito', 3: 'Indiferente',
               2: 'Insatisfeito', 1: 'Muito Insatisfeito'}
    notas = [5, 4, 3, 2, 1]

    x = np.arange(len(notas))
    largura = 0.25

    plt.figure(figsize=(10, 6))
    plt.bar(x - largura, geral.reindex(notas).values, largura, label='Geral', co
    plt.bar(x, por_sexo.loc['MASCULINO'].reindex(notas).values, largura, label='
    plt.bar(x + largura, por_sexo.loc['FEMININO'].reindex(notas).values, largura

    plt.xticks(x, [rotulos[n] for n in notas], rotation=20)
    plt.ylabel('% dos clientes')
    plt.title('Resultado da Pesquisa de Satisfação - Geral e Por Sexo')
    plt.legend()
    plt.tight_layout()
    plt.show()

    return geral, por_sexo
```

Workflow: Detalhamento das Atividades

Atividade A1: Estratificação amostral da população

Atividade A1.1: Determinar o TAMANHO IDEAL da amostra

Criar os (4) gráficos do ERRO-PADRÃO-MÉDIO (TARGET: y: **TOTAL (R$)**) das seguintes categorias:

- **ESSENTIAL**
- **VIP**
- **PRIME**
- **GERAL**

Fazer a **ANÁLISE BOOTSTRAP** para determinar:

- **Intervalo de confiança**.
- **Erro padrão**.
- **RESPONDER: Qual é o "Tamanho Ideal" da amostra ?**

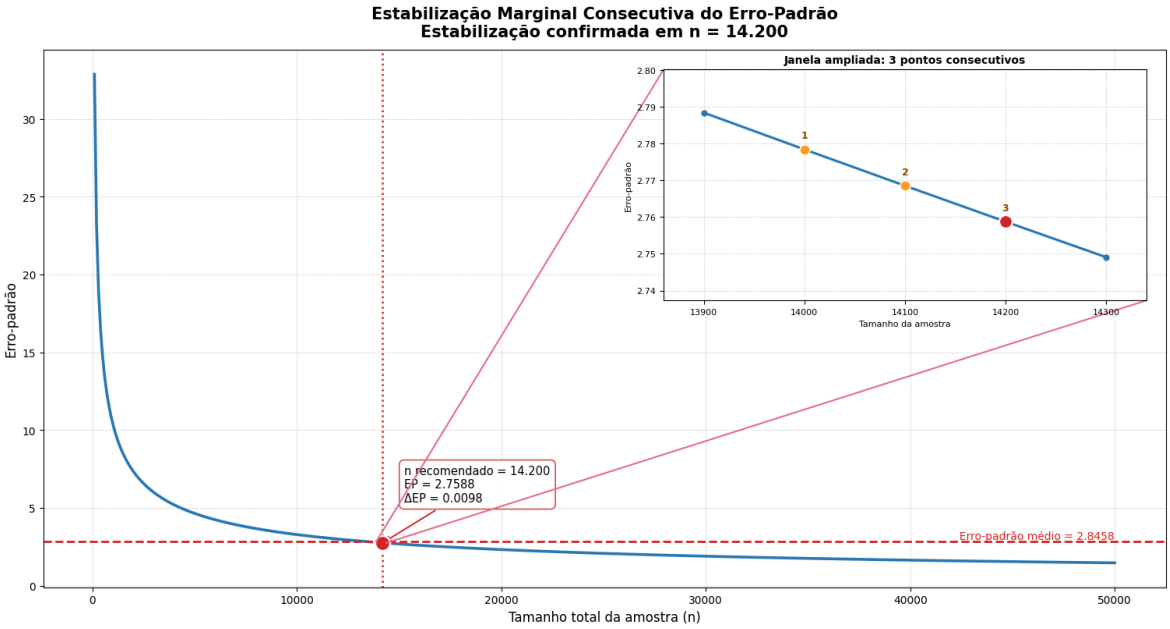
Determinação do Tamanho Ideal da Amostra em Populações Infinitas Estratificadas: Link

Usar a Função - Determinação do Tamanho Ideal da Amostra em Populações Infinitas Estratificadas, por dois(2) procedimentos a saber:

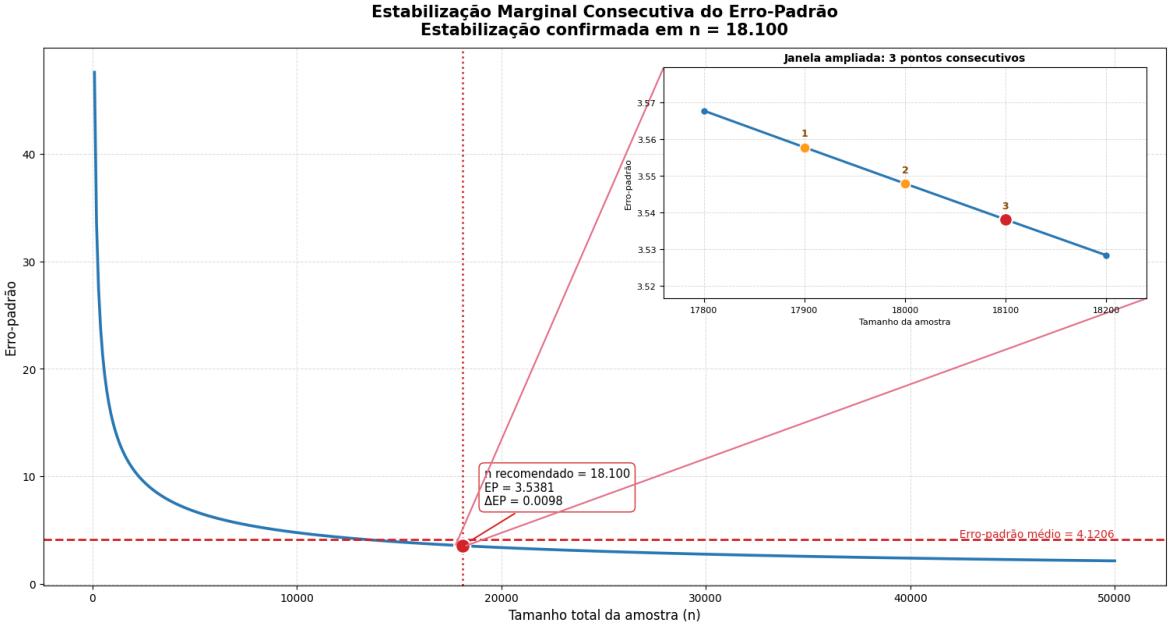
- Procedimento empírico: Estabilização marginal do erro-padrão: **optimal_sample_size**
- Procedimento teórico-empírico: Estabilização marginal do erro-padrão estratificado e otimização de Neyman: **optimal_stratified_sample_size**

In [ ]:
```
AtividadeA1_1()
```

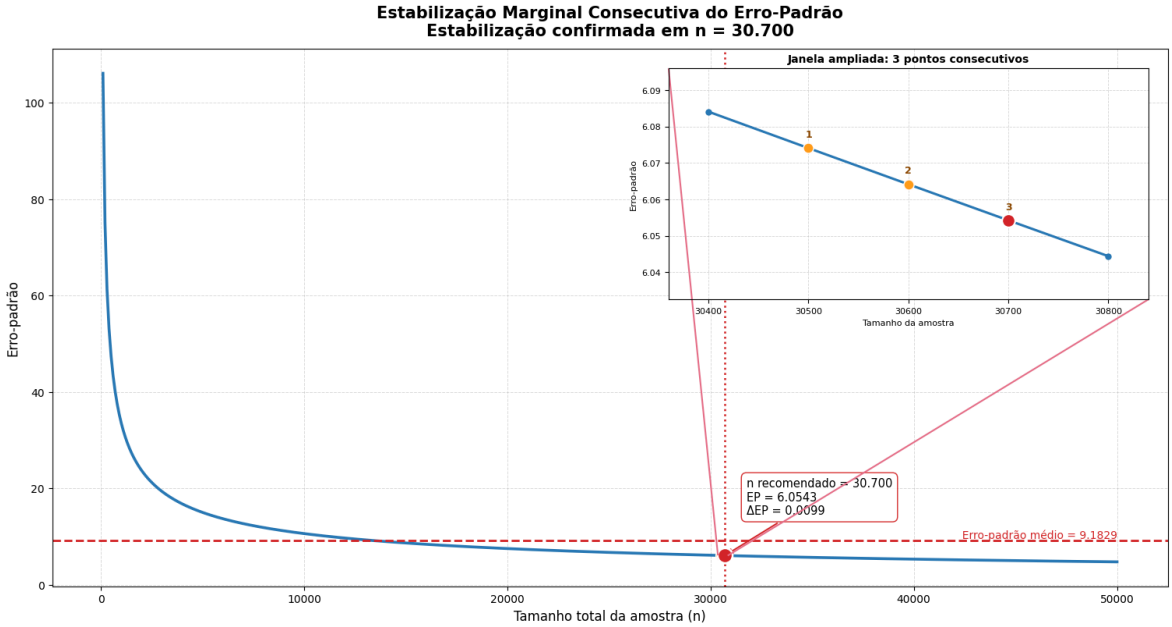
```
======================================================================
PASSO 1: Procedimento empírico (optimal_sample_size)
======================================================================
```



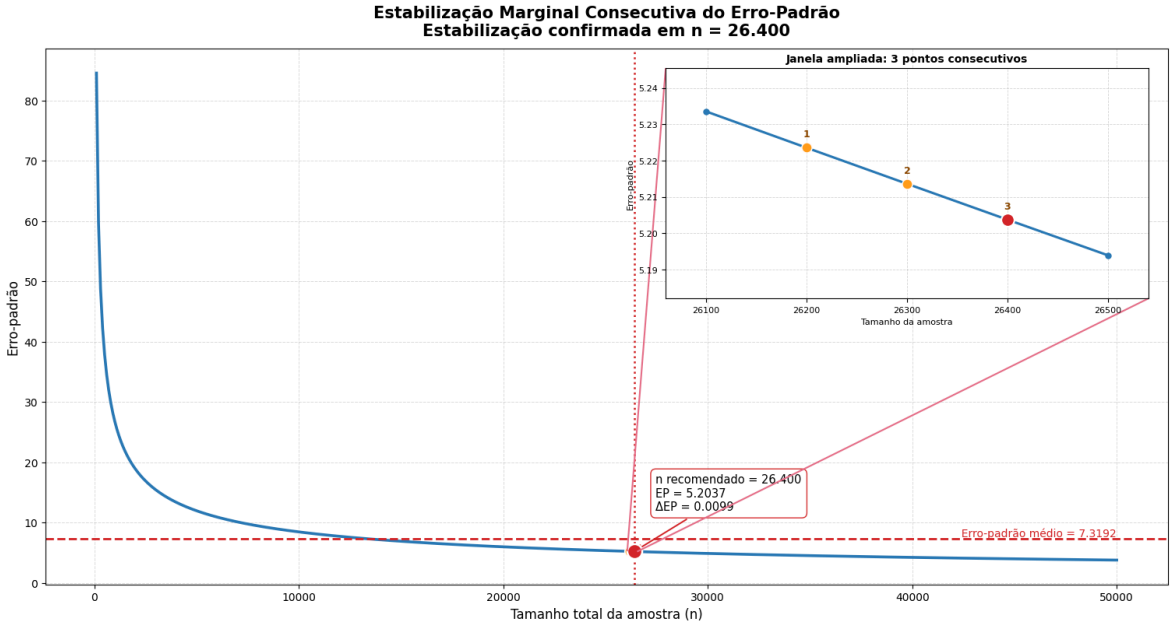

```
ESSENTIAL: n ideal = 14200, erro-padrão = R$ 2.7588
```




```
VIP: n ideal = 18100, erro-padrão = R$ 3.5381
```

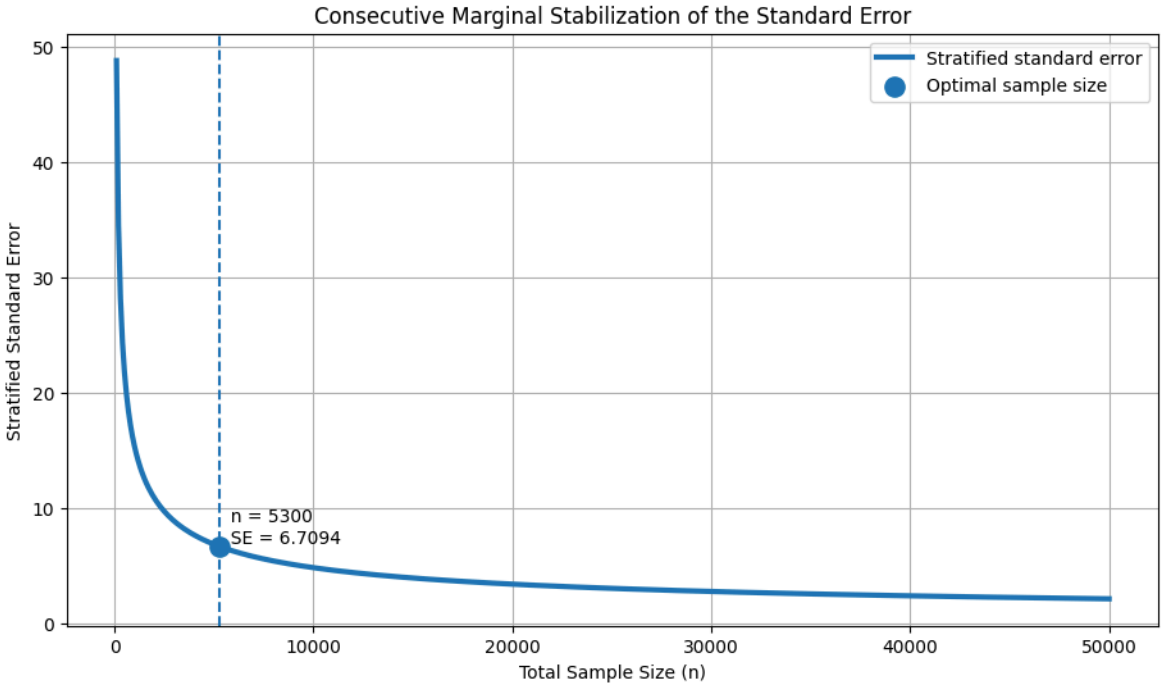



```
PRIME: n ideal = 30700, erro-padrão = R$ 6.0543
```




```
GERAL: n ideal = 26400, erro-padrão = R$ 5.2037

======================================================================
PASSO 2: Procedimento teórico-empírico (optimal_stratified_sample_size)
======================================================================
```



	Stratum	Population Weight (Wh)	Std. Deviation (Sh)	\
0	ESSENTIAL	0.595240	328.743630	
1	PRIME	0.171178	1060.791829	
2	VIP	0.233582	476.004599	

	Neyman Weight (Wh x Sh)	Allocated Sample (nh)	Allocated Sample (%)
0	195.681417	2123	0.400642
1	181.583883	1970	0.371768
2	111.186174	1206	0.227590

```
n ideal (estratificado) = 5300

======================================================================
PASSO 3: Busca do tamanho ideal por bootstrap (margem de IC <= 1%)
======================================================================
ESSENTIAL: menor n com margem <= 1% = 25,000
VIP: menor n com margem <= 1% = 10,000
PRIME: menor n com margem <= 1% = 15,000
GERAL: menor n com margem <= 1% = 40,000

Tamanho ideal da amostra adotado (maior exigência entre os grupos): 40,000

======================================================================
PASSO 4: Validação final no n adotado (n = 40,000)
======================================================================
ESSENTIAL : média real = R$    415.56 | média estimada = R$    415.60 | viés = 0.
009% | margem = ±0.78% | cobertura = 96.1%
VIP       : média real = R$    919.48 | média estimada = R$    919.58 | viés = 0.
011% | margem = ±0.51% | cobertura = 95.0%
PRIME     : média real = R$  2,155.81 | média estimada = R$  2,155.84 | viés = 0.
001% | margem = ±0.48% | cobertura = 97.0%
GERAL     : média real = R$    831.16 | média estimada = R$    831.29 | viés = 0.
016% | margem = ±1.00% | cobertura = 96.0%

RESPOSTA FINAL: o tamanho ideal da amostra é 40,000, validado como não-viesado, p
reciso e estatisticamente confiável em todos os grupos analisados.
```

Out[ ]:
```
{'estratificado':      Stratum  Population Weight (Wh)  Std. Deviation (Sh)  \
  0  ESSENTIAL                0.595240           328.743630
  1      PRIME                0.171178          1060.791829
  2        VIP                0.233582           476.004599

     Neyman Weight (Wh x Sh)  Allocated Sample (nh)  Allocated Sample (%)
  0               195.681417                   2123              0.400642
  1               181.583883                   1970              0.371768
  2               111.186174                   1206              0.227590  ,
  {'optimal_n': 5300,
   'optimal_standard_error': 6.709396995239152,
   'sample_sizes': array([  100,   200,   300,   400,   500,   600,   700,   800,   900,
          1000,  1100,  1200,  1300,  1400,  1500,  1600,  1700,  1800,
          1900,  2000,  2100,  2200,  2300,  2400,  2500,  2600,  2700,
          2800,  2900,  3000,  3100,  3200,  3300,  3400,  3500,  3600,
          3700,  3800,  3900,  4000,  4100,  4200,  4300,  4400,  4500,
          4600,  4700,  4800,  4900,  5000,  5100,  5200,  5300,  5400,
          5500,  5600,  5700,  5800,  5900,  6000,  6100,  6200,  6300,
          6400,  6500,  6600,  6700,  6800,  6900,  7000,  7100,  7200,
          7300,  7400,  7500,  7600,  7700,  7800,  7900,  8000,  8100,
          8200,  8300,  8400,  8500,  8600,  8700,  8800,  8900,  9000,
          9100,  9200,  9300,  9400,  9500,  9600,  9700,  9800,  9900,
         10000, 10100, 10200, 10300, 10400, 10500, 10600, 10700, 10800,
         10900, 11000, 11100, 11200, 11300, 11400, 11500, 11600, 11700,
         11800, 11900, 12000, 12100, 12200, 12300, 12400, 12500, 12600,
         12700, 12800, 12900, 13000, 13100, 13200, 13300, 13400, 13500,
         13600, 13700, 13800, 13900, 14000, 14100, 14200, 14300, 14400,
         14500, 14600, 14700, 14800, 14900, 15000, 15100, 15200, 15300,
         15400, 15500, 15600, 15700, 15800, 15900, 16000, 16100, 16200,
         16300, 16400, 16500, 16600, 16700, 16800, 16900, 17000, 17100,
         17200, 17300, 17400, 17500, 17600, 17700, 17800, 17900, 18000,
         18100, 18200, 18300, 18400, 18500, 18600, 18700, 18800, 18900,
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         19900, 20000, 20100, 20200, 20300, 20400, 20500, 20600, 20700,
         20800, 20900, 21000, 21100, 21200, 21300, 21400, 21500, 21600,
         21700, 21800, 21900, 22000, 22100, 22200, 22300, 22400, 22500,
         22600, 22700, 22800, 22900, 23000, 23100, 23200, 23300, 23400,
         23500, 23600, 23700, 23800, 23900, 24000, 24100, 24200, 24300,
         24400, 24500, 24600, 24700, 24800, 24900, 25000, 25100, 25200,
         25300, 25400, 25500, 25600, 25700, 25800, 25900, 26000, 26100,
         26200, 26300, 26400, 26500, 26600, 26700, 26800, 26900, 27000,
         27100, 27200, 27300, 27400, 27500, 27600, 27700, 27800, 27900,
         28000, 28100, 28200, 28300, 28400, 28500, 28600, 28700, 28800,
         28900, 29000, 29100, 29200, 29300, 29400, 29500, 29600, 29700,
         29800, 29900, 30000, 30100, 30200, 30300, 30400, 30500, 30600,
         30700, 30800, 30900, 31000, 31100, 31200, 31300, 31400, 31500,
         31600, 31700, 31800, 31900, 32000, 32100, 32200, 32300, 32400,
         32500, 32600, 32700, 32800, 32900, 33000, 33100, 33200, 33300,
         33400, 33500, 33600, 33700, 33800, 33900, 34000, 34100, 34200,
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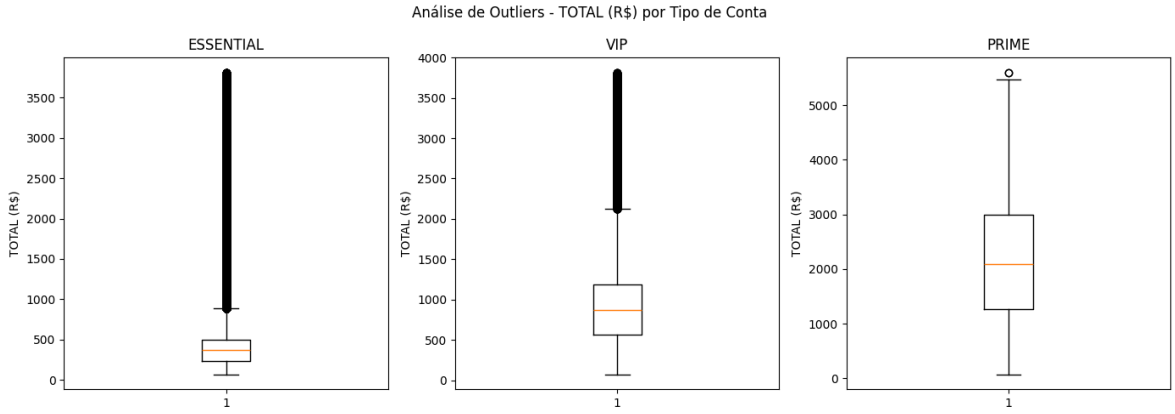
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  'vies_%': np.float64(0.01086101843829927),
  'margem_erro_%': np.float64(0.5073344894678139),
  'cobertura_%': 95.0},
 'PRIME': {'media_real': np.float64(2155.8116608682503),
  'media_estimada': np.float64(2155.842724574),
  'vies_%': np.float64(0.0014409285520374034),
  'margem_erro_%': np.float64(0.4822202290288303),
  'cobertura_%': 97.0},
 'GERAL': {'media_real': np.float64(831.1605738660716),
  'media_estimada': np.float64(831.289617447),
  'vies_%': np.float64(0.01552571007166055),
```

```
    'margem_erro_%': np.float64(0.9969104565036616),
    'cobertura_%': 96.0}}}
```

In [ ]:
```
help(AtividadeA1_1)
```

Atividade A1.2: Analisar os outliers

In [ ]:
```
AtividadeA1_2()
```




```
     Tipo      Q1      Q3     IQR  Limite inferior  Limite superior  Qtd outliers
% outliers
ESSENTIAL  239.16  498.55  259.39                0           887.64         24371
3.66
      VIP  561.89 1187.58  625.69                0          2126.12          7087
2.71
    PRIME 1267.32 2990.98 1723.66                0          5576.48             2
0.00

DECISÃO: Outliers MANTIDOS na base.
Justificativa: representam gastos legítimos de clientes de alto
consumo, não erros de coleta ou digitação. Removê-los distorceria
o comportamento real de consumo de cada tipo de conta.
```

Out[ ]:

	Tipo	Q1	Q3	IQR	Limite inferior	Limite superior	Qtd outliers	% outliers
0	ESSENTIAL	239.16	498.55	259.39	0	887.64	24371	3.66
1	VIP	561.89	1187.58	625.69	0	2126.12	7087	2.71
2	PRIME	1267.32	2990.98	1723.66	0	5576.48	2	0.00

Atividade A1.3: Testes de Hipóteses (Link)

A1.3.1: Teste de Normalidade

Nesta ETAPA, você deve fazer o **TESTE DE HIPÓTESE DE NORMALIDADE: Anderson-Darling' Test (AD test)** sobre:

- SAMPLE 1: ESSENTIAL (n = 40000)
- SAMPLE 2: PRIME (n = 40000)

In [ ]:
```
AtividadeA1_3_1()
```

```
SAMPLE 1 (ESSENTIAL) - DISTRIBUIÇÃO AMOSTRAL: stat = 3556.771
  Intervalo de Confiança (15.0%): Provavelmente NÃO Gaussiana.
  Intervalo de Confiança (10.0%): Provavelmente NÃO Gaussiana.
  Intervalo de Confiança (5.0%): Provavelmente NÃO Gaussiana.
  Intervalo de Confiança (2.5%): Provavelmente NÃO Gaussiana.
  Intervalo de Confiança (1.0%): Provavelmente NÃO Gaussiana.

SAMPLE 2 (PRIME) - DISTRIBUIÇÃO AMOSTRAL: stat = 253.174
  Intervalo de Confiança (15.0%): Provavelmente NÃO Gaussiana.
  Intervalo de Confiança (10.0%): Provavelmente NÃO Gaussiana.
  Intervalo de Confiança (5.0%): Provavelmente NÃO Gaussiana.
  Intervalo de Confiança (2.5%): Provavelmente NÃO Gaussiana.
  Intervalo de Confiança (1.0%): Provavelmente NÃO Gaussiana.
```

Conclusão

ESSENTIAL e PRIME foram escolhidos por representarem os extremos de gasto e variabilidade do clube (A1.1), tornando a comparação estatística mais informativa que entre grupos intermediários.

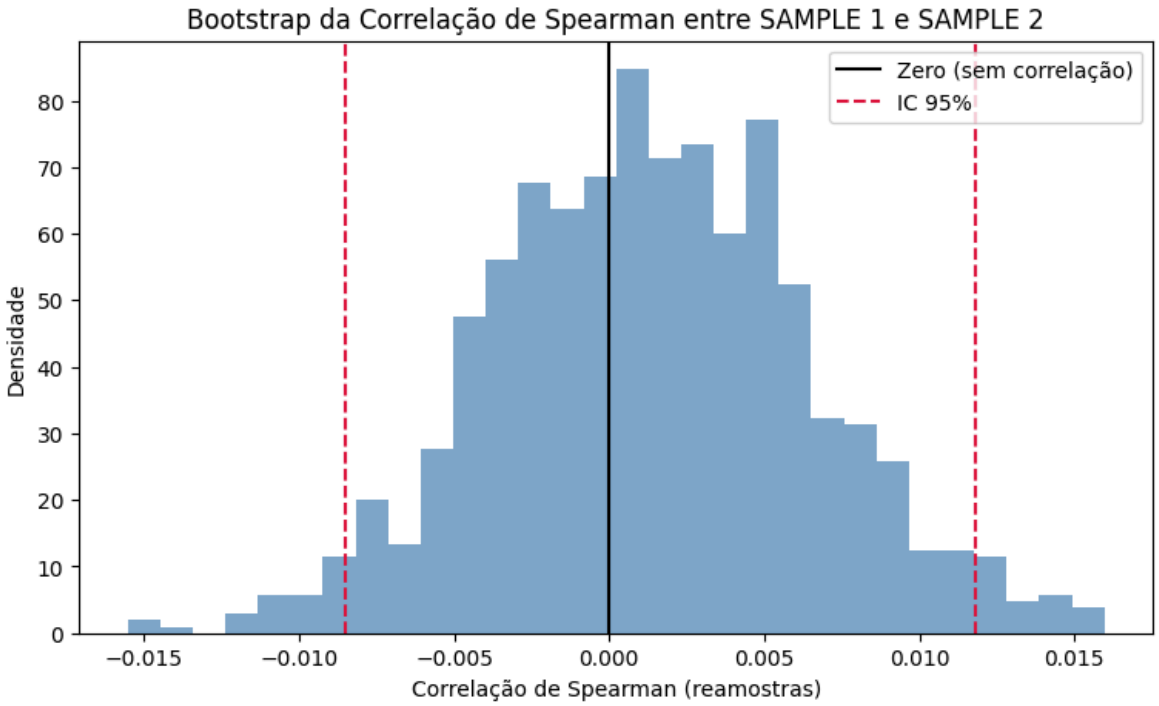
Nenhuma das duas amostras segue distribuição normal, em nenhum dos níveis de confiança testados (15% a 1%).

A1.3.2: Teste de Independência

Nesta ETAPA, você deve fazer o **TESTE DE ANOVA (+ Gráfico Bootstrap) sobre Testes de Independência: p_value** sobre:

In [ ]:
```
AtividadeA1_3_2()
```

```
Pearson: stat = -0.0038, p-valor = 0.4483
Spearman: stat = 0.0015, p-valor = 0.7666
Kendall: stat = 0.0010, p-valor = 0.7671
```




```
IC 95% (bootstrap) da correlação de Spearman: ConfidenceInterval(low=np.float64(-
0.008500684963637482), high=np.float64(0.011812939497938322))
```

Out[ ]:
```
({'Pearson': PearsonRResult(statistic=np.float64(-0.003790998198850531), pvalue
=np.float64(0.4483440287979301)),
  'Spearman': SignificanceResult(statistic=np.float64(0.0014841188410606426), p
value=np.float64(0.7666080157448258)),
  'Kendall': SignificanceResult(statistic=np.float64(0.0009874997113243746), pv
alue=np.float64(0.767052285134972))},
 BootstrapResult(confidence_interval=ConfidenceInterval(low=np.float64(-0.00850
0684963637482), high=np.float64(0.011812939497938322)), bootstrap_distribution=
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tandard_error=np.float64(0.005103534497911512)))
```

Conclusão

- **SAMPLE 1 e SAMPLE 2 são independentes**: os três testes de correlação (Pearson, Spearman, Kendall) retornaram p-valor >> 0,05, não havendo evidência de relação estatística entre as amostras.
- **BEST FIT**: Spearman (ou Kendall), pois não exigem normalidade dos dados — diferente de Pearson, que assume relação linear e distribuição aproximadamente normal, violada conforme A1.3.1.

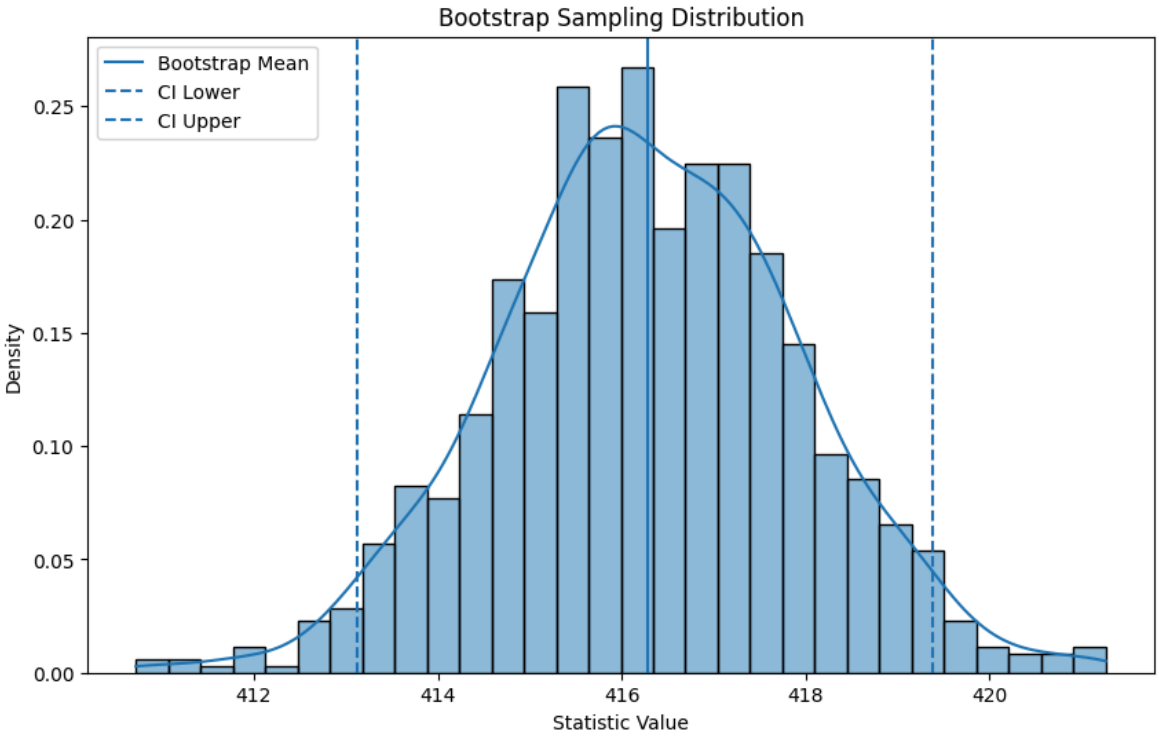
A1.3.3: Teste de Distribuição

Nesta ETAPA, você deve fazer o **TESTE DE ANOVA (+ Gráfico Bootstrap) sobre Teste de Mesma Distribuição: p_value**

In [ ]:
```
AtividadeA1_3_3()
```

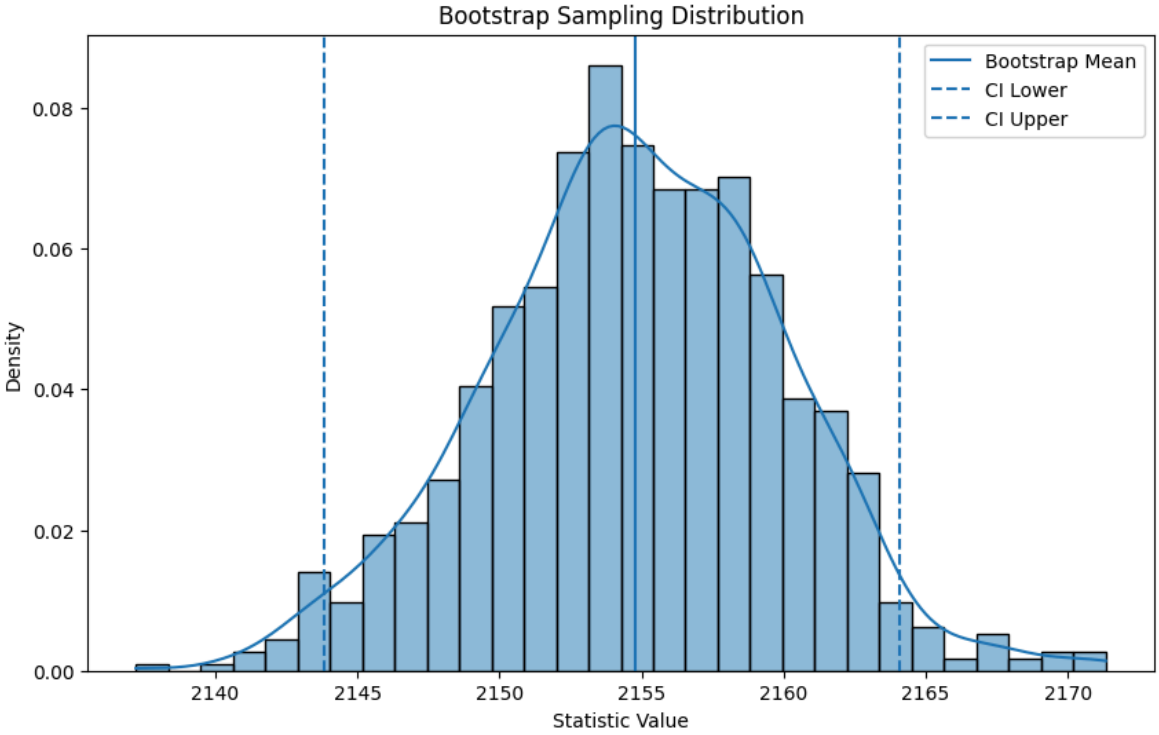
```
ttest_ind (paramétrico): stat = -313.8190, p-valor = 0.000000
f_oneway (paramétrico): stat = 98482.3598, p-valor = 0.000000
mannwhitneyu (não-paramétrico): stat = 61990797.0000, p-valor = 0.000000
kruskal (não-paramétrico): stat = 51061.0102, p-valor = 0.000000

--- Gráfico Bootstrap: SAMPLE 1 (ESSENTIAL) ---
```



CONFIDENCE INTERVAL
LOW : 413.1207
HIGH : 419.3846
SE : 1.6379

--- Gráfico Bootstrap: SAMPLE 2 (PRIME) ---



CONFIDENCE INTERVAL
LOW : 2143.8302
HIGH : 2164.0817
SE : 5.1532

Out[ ]:
```
({'ttest_ind (paramétrico)': TtestResult(statistic=np.float64(-313.818992169815), pvalue=np.float64(0.0), df=np.float64(79998.0)),
  'f_oneway (paramétrico)': F_onewayResult(statistic=np.float64(98482.359846478 2), pvalue=np.float64(0.0)),
  'mannwhitneyu (não-paramétrico)': MannwhitneyuResult(statistic=np.float64(619 90797.0), pvalue=np.float64(0.0)),
  'kruskal (não-paramétrico)': KruskalResult(statistic=np.float64(51061.0102309 7901), pvalue=np.float64(0.0))},
 (BootstrapResult(confidence_interval=ConfidenceInterval(low=np.float64(413.120 68489999996), high=np.float64(419.38458086249994)), bootstrap_distribution=arra y([416.9094245 , 419.37449425, 417.257711  , 416.581837  ,
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        2162.715715  , 2156.0221485 , 2155.3102    , 2153.85900275]), standard
_error=np.float64(5.15317041536782))))
```

Conclusão

- **SAMPLE 1 e SAMPLE 2 não têm a mesma distribuição**: todos os 4 testes (paramétricos e não-paramétricos) rejeitam essa hipótese com **p-valor ≈ 0**.
- **BEST FIT**: Kruskal-Wallis (não-paramétrico), consistente com a não-normalidade confirmada na A1.3.1. Resultado concorda com o esperado, já que ESSENTIAL e PRIME representam perfis de gasto opostos por desenho do negócio.

A1.3.4: Teste das Médias Amostrais

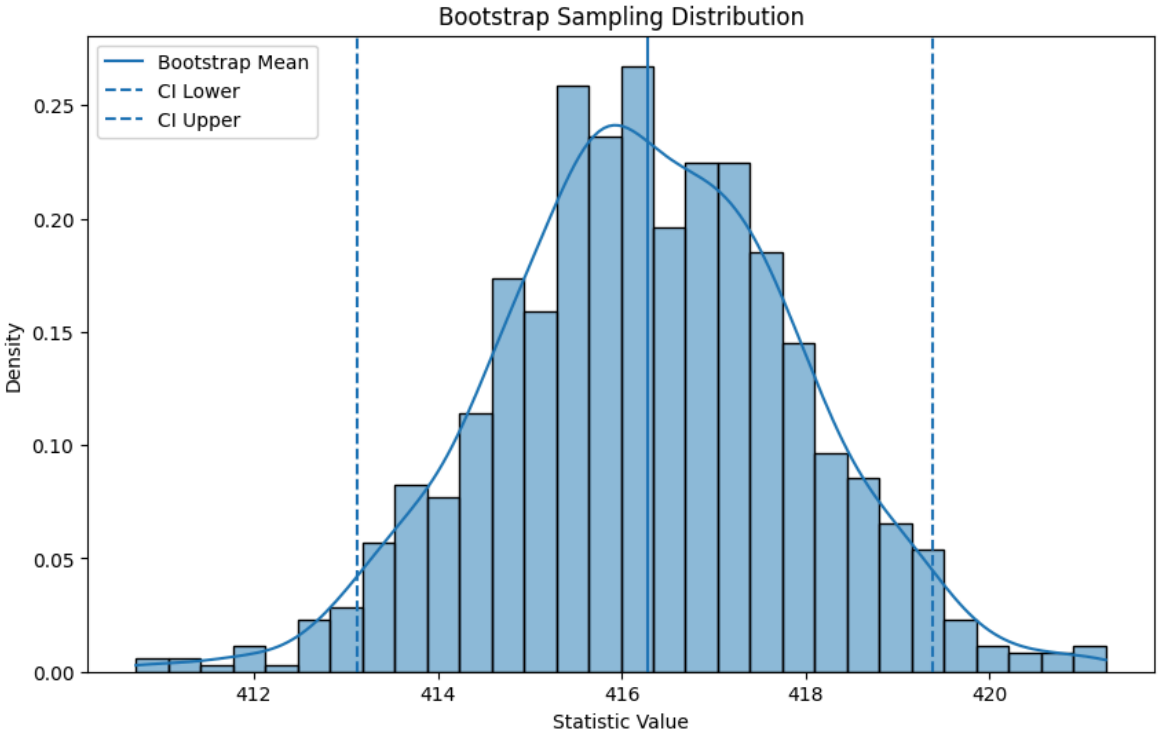
Nesta ETAPA, você deve fazer o **TESTE DE ANOVA (+ Gráfico Bootstrap) sobre Teste das Médias Amostrais: p_value** sobre:

- SAMPLE 1 e SAMPLE 2
- RESPONDER: Qual o BEST FIT ?

In [ ]:
```
AtividadeA1_3_4()
```

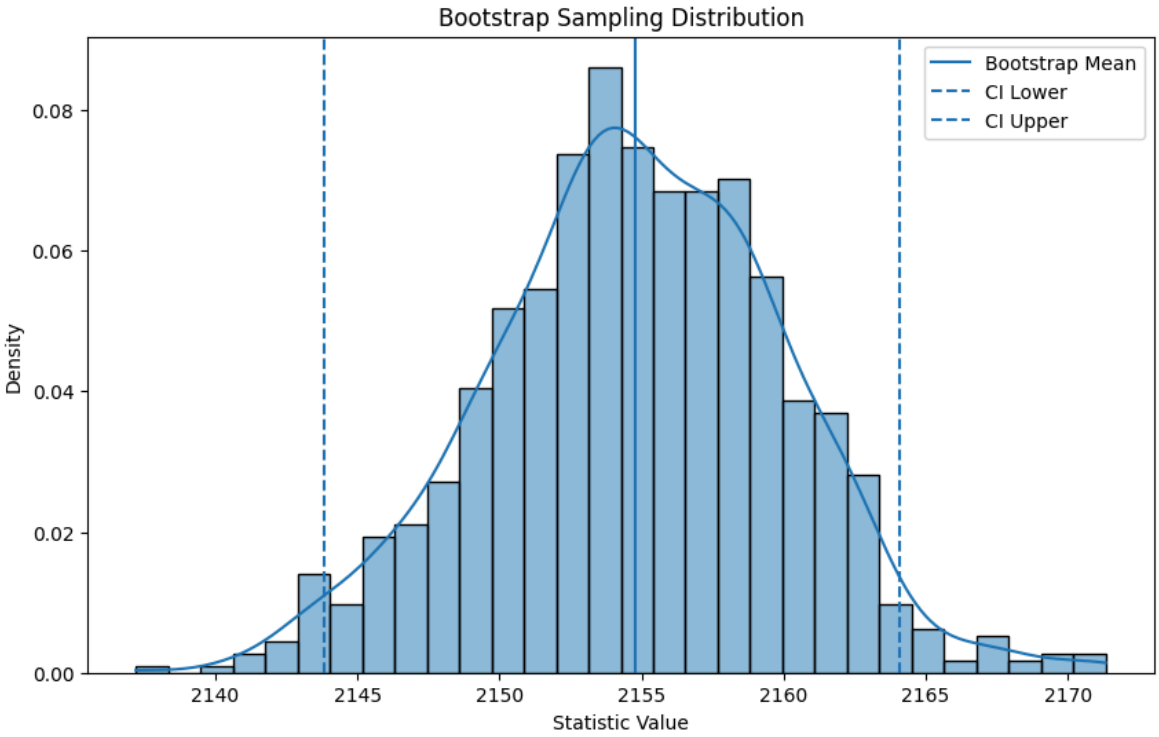
```
ttest_ind: stat = -313.8190, p-valor = 0.000000
Média SAMPLE 1 (ESSENTIAL): R$ 416.21
Média SAMPLE 2 (PRIME): R$ 2,154.59

--- Gráfico Bootstrap: SAMPLE 1 (ESSENTIAL) ---
```




```
CONFIDENCE INTERVAL
LOW  : 413.1207
HIGH : 419.3846
SE   : 1.6379

--- Gráfico Bootstrap: SAMPLE 2 (PRIME) ---
```




```
CONFIDENCE INTERVAL
LOW  : 2143.8302
HIGH : 2164.0817
SE   : 5.1532
```

Out[ ]:
```
(TtestResult(statistic=np.float64(-313.818992169815), pvalue=np.float64(0.0), df=np.float64(79998.0)),
 (BootstrapResult(confidence_interval=ConfidenceInterval(low=np.float64(413.12068489999996), high=np.float64(419.38458086249994)), bootstrap_distribution=array([416.9094245 , 419.37449425, 417.257711  , 416.581837  ,
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```
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```

```
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            2154.91896225, 2156.346916  , 2159.2770925 , 2159.536113  ,
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            2155.281301  , 2151.46804525, 2148.34428875, 2163.2477085 ,
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            2156.51200875, 2148.6994485 , 2153.4013425 , 2150.5838925 ,
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            2154.18472975, 2150.085857  , 2162.14480525, 2158.31449125,
            2157.6414065 , 2154.5421835 , 2154.0496345 , 2150.5347415 ,
            2155.9572695 , 2161.424408  , 2160.789377  , 2159.66806575,
            2154.4406985 , 2163.2355015 , 2148.529634  , 2154.291188  ,
            2154.36320875, 2148.839813  , 2152.86399475, 2163.666488  ,
            2162.715715  , 2156.0221485 , 2155.3102    , 2153.85900275]), standard
_error=np.float64(5.15317041536782))))
```

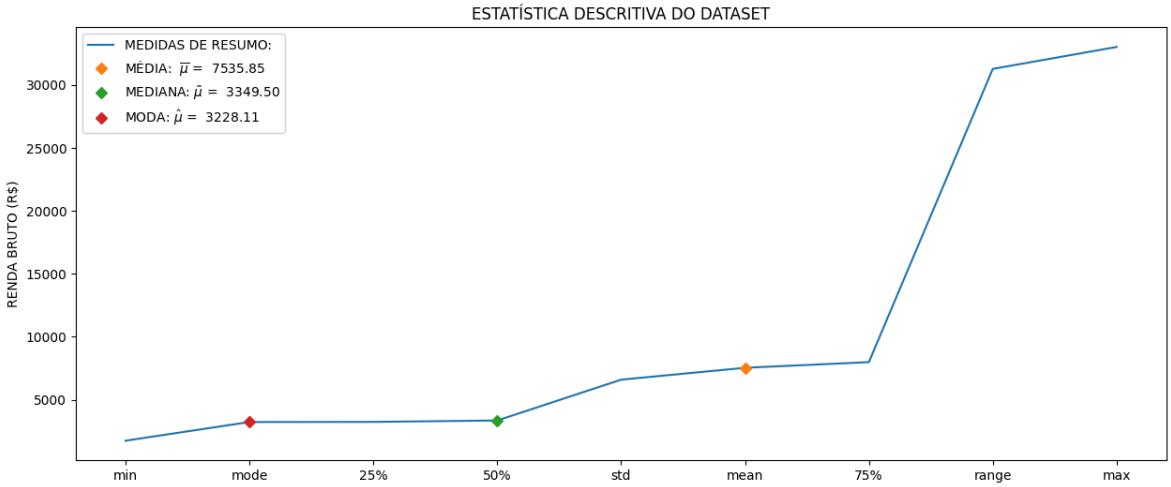
Conclusão

- **SAMPLE 1 e SAMPLE 2 não têm as mesmas médias amostrais**: o ttest_ind rejeita a igualdade com p-valor ≈ 0 (stat = -313,82). Os IC 95% bootstrap de cada amostra (ESSENTIAL: `R$ 413,12` a `R$ 419,38` ; PRIME: `R$ 2.143,83` a `R$ 2.164,08`) não se sobrepõem, confirmando visualmente a diferença.
- **Resultado esperado e coerente:** ESSENTIAL e PRIME são, por desenho do negócio, os extremos de gasto do clube — a diferença de média (`R$ 416` vs `R$ 2.155`) reflete exatamente esse perfil.

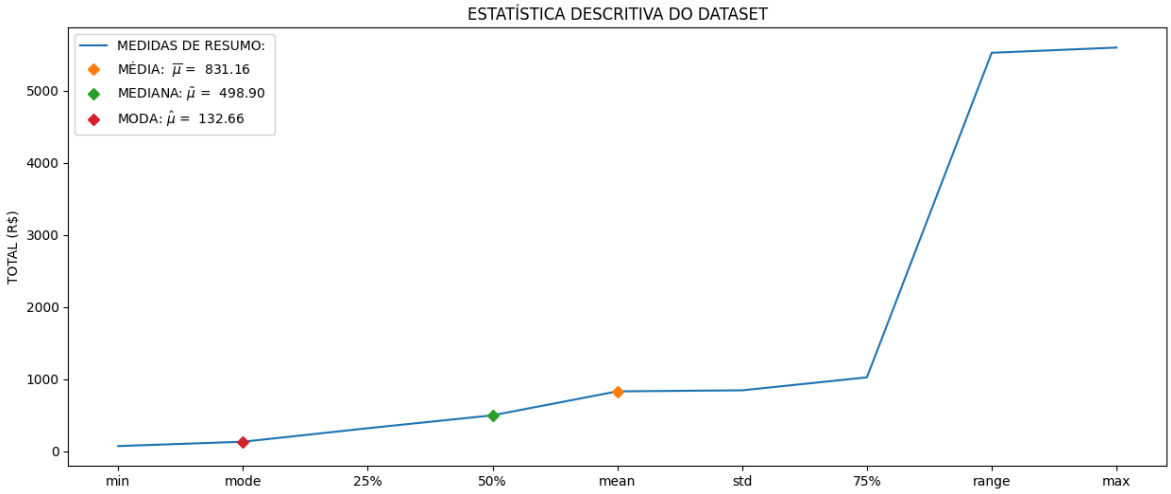
Atividade A2 (Link): Estatística descritiva

Atividade A2.1: Analisar as variáveis Quantitativas

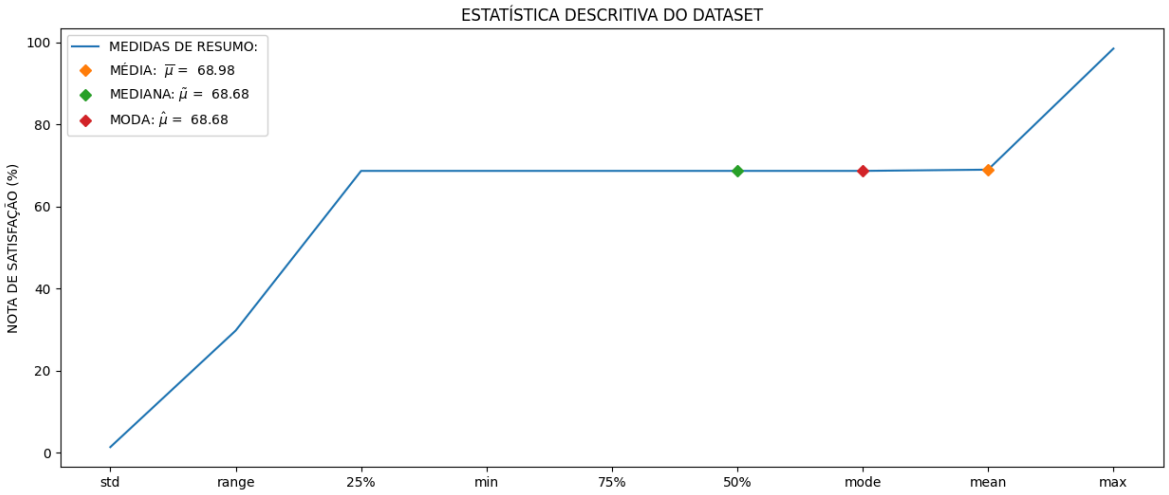
In [ ]:
```
AtividadeA2_1()
```




```
ESTATÍSTICA DESCRITIVA DO DATASET:
min       1744.17
mode      3228.11
25%       3237.52
50%       3349.50
std       6591.67
mean      7535.85
75%       7984.86
range    31279.79
max      33023.96
Name: RENDA BRUTO (R$), dtype: float64
```




```
ESTATÍSTICA DESCRITIVA DO DATASET:
min        72.48
mode      132.66
25%       319.88
50%       498.90
mean      831.16
std       845.50
75%      1025.26
range    5525.32
max      5597.80
Name: TOTAL (R$), dtype: float64
```


ESTATÍSTICA DESCRITIVA DO DATASET
MEDIDAS DE RESUMO:
MÉDIA: $\overline{\mu}$ = 68.98
MEDIANA: $\tilde{\mu}$ = 68.68
MODA: $\hat{\mu}$ = 68.68
NOTA DE SATISFAÇÃO (%)
100
80
60
40
20
0
std
range
25%
min
75%
50%
mode
mean
max

```
ESTATÍSTICA DESCRITIVA DO DATASET:
std       1.40
range    29.79
25%      68.68
min      68.68
75%      68.68
50%      68.68
mode     68.68
mean     68.98
max      98.48
Name: NOTA DE SATISFAÇÃO (%), dtype: float64
PERFIL QUÍMICO DO VINHO (consolidado):
       MEAN: alcohol  MEAN: malic_acid  MEAN: ash  MEAN: alcalinity_of_ash  \
25%            12.45              1.58       2.18                    17.25
50%            13.00              2.34       2.37                    19.49
75%            13.55              3.09       2.55                    21.74
max            16.87              7.60       3.68                    35.15
mean           13.00              2.34       2.37                    19.49
min             8.93             -3.35       0.88                     4.03
mode            8.93             -3.35       0.88                     4.03
range           7.94             10.96       2.80                    31.11
std             0.81              1.11       0.27                     3.33

       MEAN: magnesium  MEAN: total_phenols  MEAN: flavanoids  \
25%              90.12                 1.87              1.36
50%              99.75                 2.29              2.03
75%             109.36                 2.71              2.70
max             170.05                 5.45              6.93
mean             99.74                 2.29              2.03
min              31.99                -1.23             -2.91
mode             31.99                -1.23             -2.91
range           138.06                 6.68              9.84
std              14.24                 0.62              0.99

       MEAN: nonflavanoid_phenols  MEAN: proanthocyanins  \
25%                          0.28                   1.21
50%                          0.36                   1.59
75%                          0.45                   1.98
max                          1.04                   4.25
mean                         0.36                   1.59
min                         -0.25                  -1.18
mode                        -0.25                  -1.18
range                        1.29                   5.43
std                          0.12                   0.57

       MEAN: color_intensity  MEAN: hue  MEAN: od280/od315_of_diluted_wines  \
25%                     3.50       0.80                                2.13
50%                     5.06       0.96                                2.61
75%                     6.62       1.11                                3.09
max                    16.61       2.09                                5.95
mean                    5.06       0.96                                2.61
min                    -5.91      -0.20                               -0.90
mode                   -5.91      -0.20                               -0.90
range                  22.51       2.29                                6.85
std                     2.31       0.23                                0.71

       MEAN: proline
25%           535.56
50%           746.69
75%           958.52
```

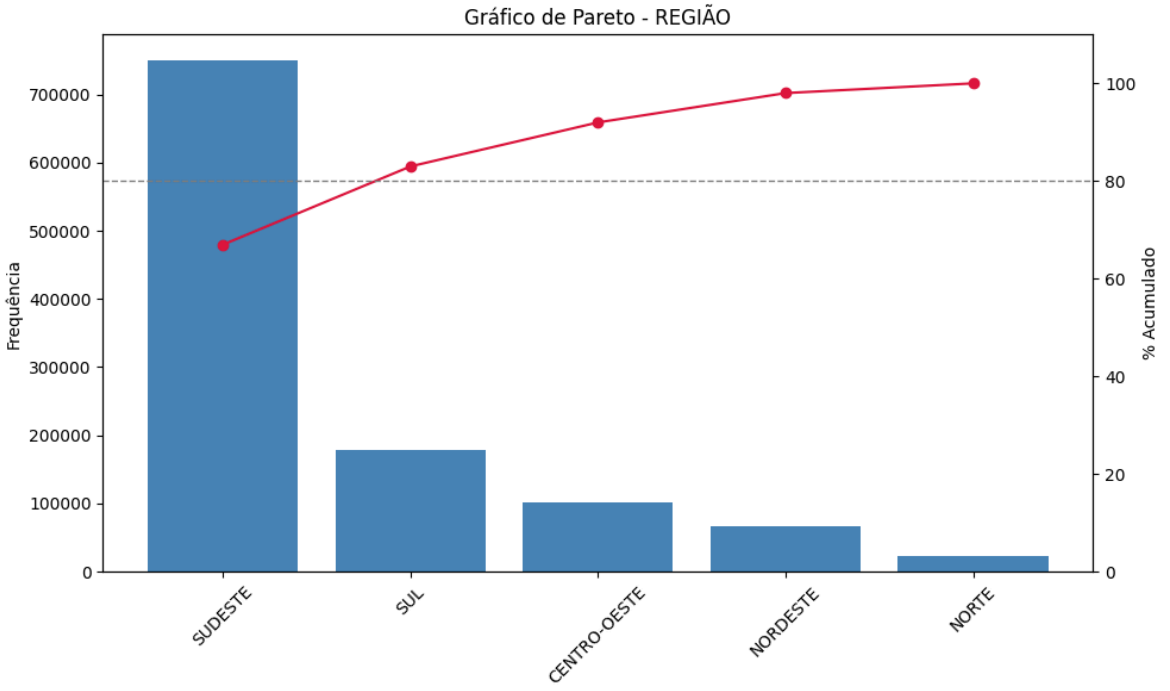
```
max          2321.29
mean          746.99
min          -754.41
mode         -754.41
range        3075.71
std           313.81
```

Out[ ]:

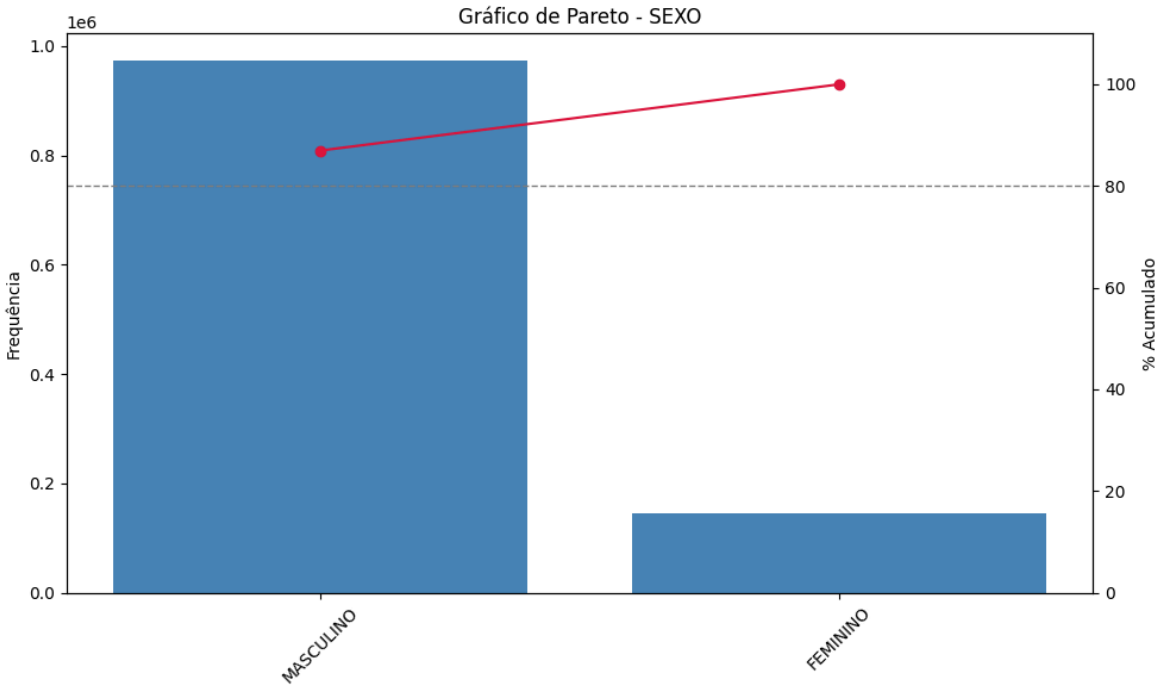
	MEAN: alcohol	MEAN: malic_acid	MEAN: ash	MEAN: alcalinity_of_ash	MEAN: magnesium	MEAN: total_phenols	MEA flavanc
25%	12.45	1.58	2.18	17.25	90.12	1.87	
50%	13.00	2.34	2.37	19.49	99.75	2.29	
75%	13.55	3.09	2.55	21.74	109.36	2.71	
max	16.87	7.60	3.68	35.15	170.05	5.45	
mean	13.00	2.34	2.37	19.49	99.74	2.29	
min	8.93	-3.35	0.88	4.03	31.99	-1.23	
mode	8.93	-3.35	0.88	4.03	31.99	-1.23	
range	7.94	10.96	2.80	31.11	138.06	6.68	
std	0.81	1.11	0.27	3.33	14.24	0.62	

Atividade A2.2: Analisar as variáveis Qualitativas

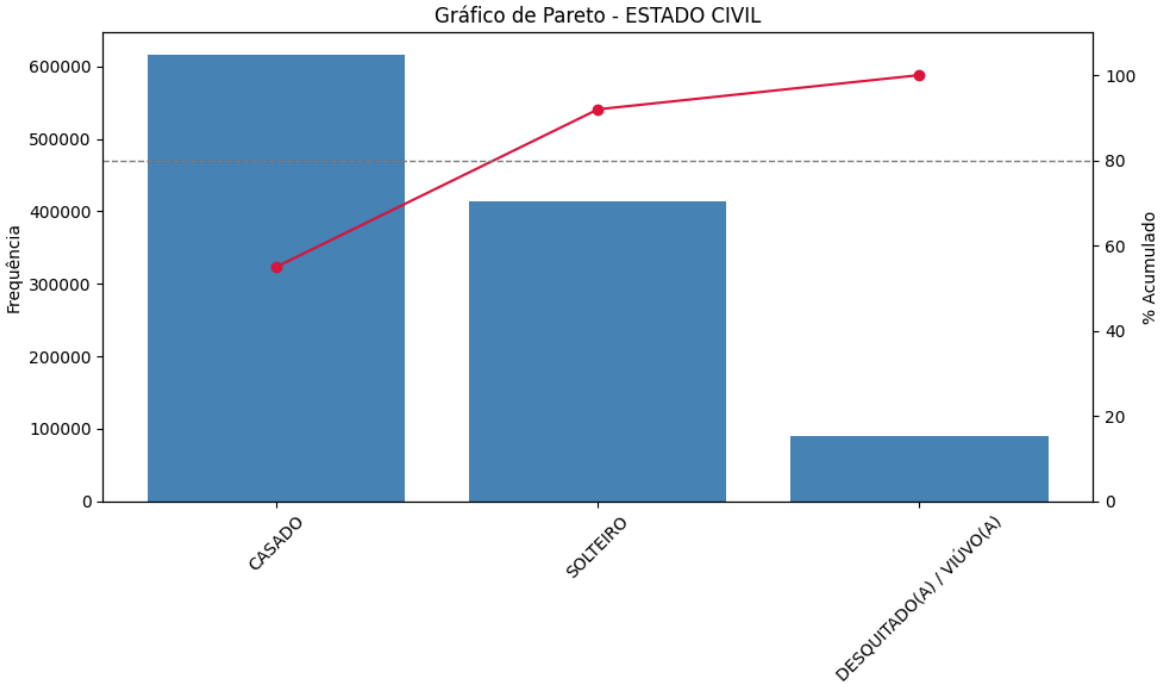
In [ ]:
```
AtividadeA2_2()
```



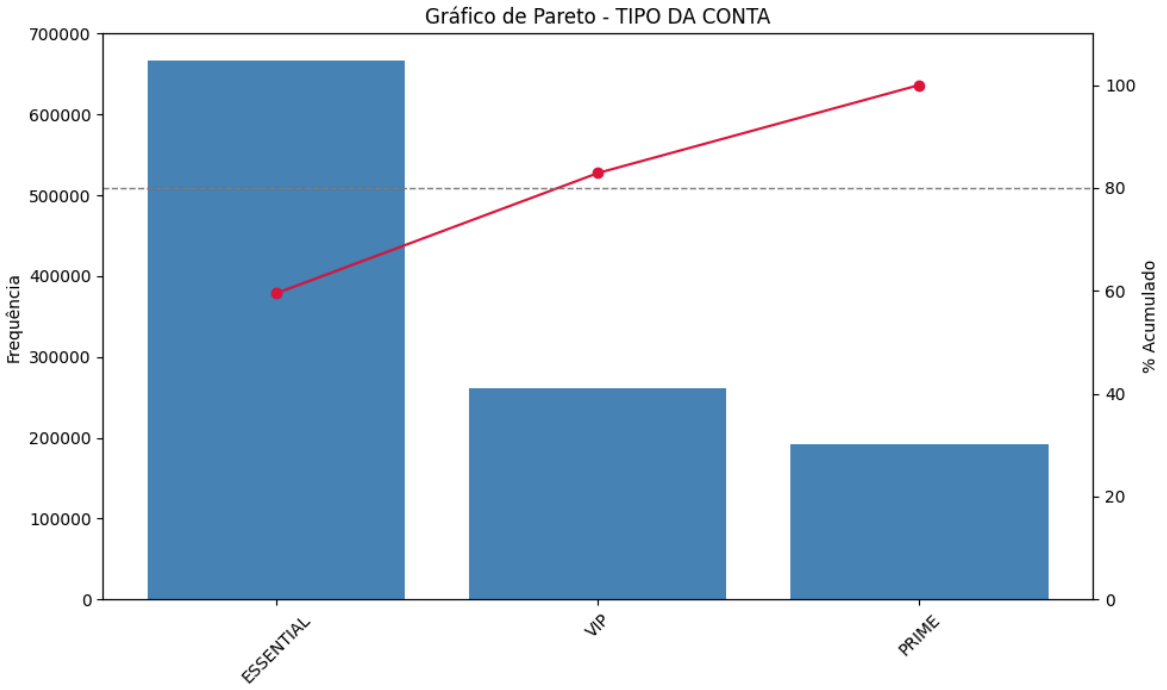

```
REGIÃO: 5 categorias; 2 delas já somam 80% do total.
```



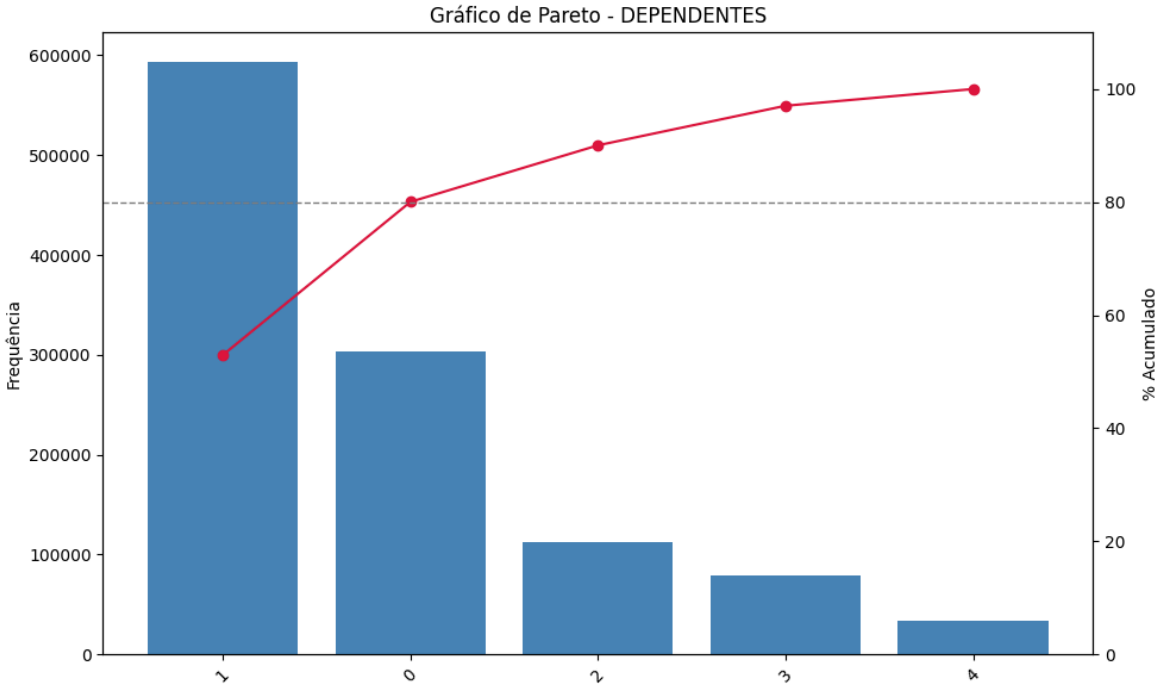
SEXO: 2 categorias; 1 delas já somam 80% do total.



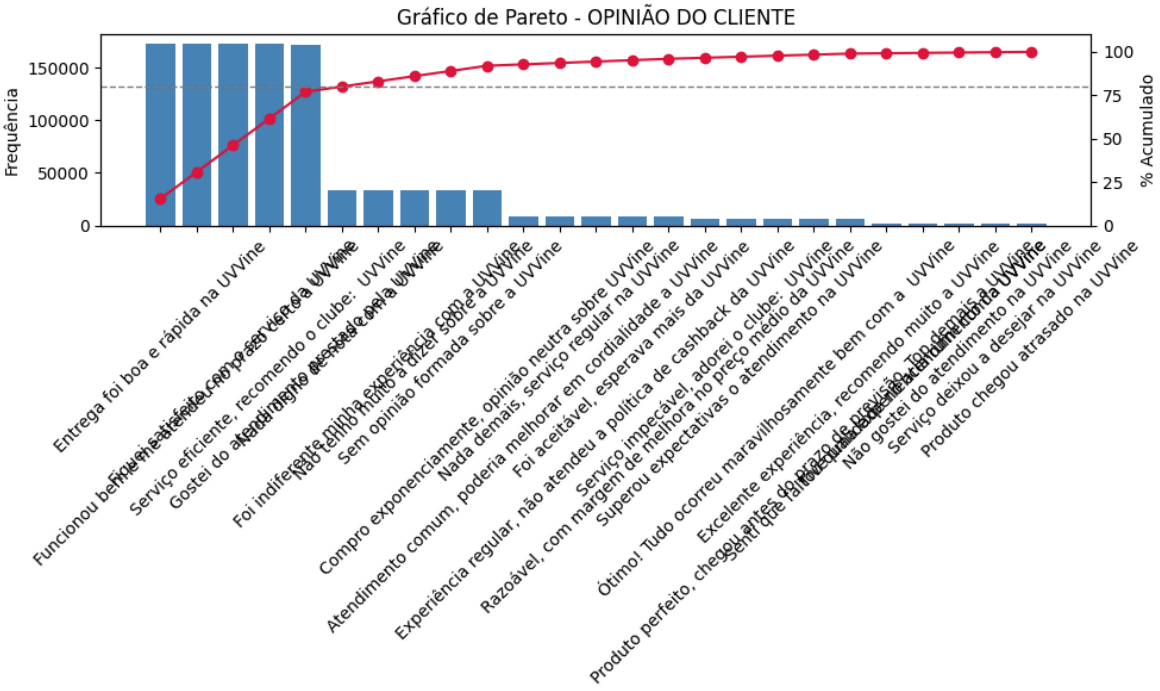
ESTADO CIVIL: 3 categorias; 2 delas já somam 80% do total.



TIPO DA CONTA: 3 categorias; 2 delas já somam 80% do total.



DEPENDENTES: 5 categorias; 2 delas já somam 80% do total.



OPINIÃO DO CLIENTE: 25 categorias; 6 delas já somam 80% do total.

Out[ ]:
```
{'REGIÃO': (REGIÃO
 SUDESTE         750186
 SUL             179381
 CENTRO-OESTE    100870
 NORDESTE         67294
 NORTE            22269
 Name: count, dtype: int64,
 REGIÃO
 SUDESTE          66.980893
 SUL              82.997054
 CENTRO-OESTE     92.003304
 NORDESTE         98.011696
 NORTE           100.000000
 Name: count, dtype: float64),
'SEXO': (SEXO
 MASCULINO    974185
 FEMININO     145815
 Name: count, dtype: int64,
 SEXO
 MASCULINO     86.980804
 FEMININO     100.000000
 Name: count, dtype: float64),
'ESTADO CIVIL': (ESTADO CIVIL
 CASADO                      616145
 SOLTEIRO                    413995
 DESQUITADO(A) / VIÚVO(A)     89860
 Name: count, dtype: int64,
 ESTADO CIVIL
 CASADO                       55.012946
 SOLTEIRO                     91.976786
 DESQUITADO(A) / VIÚVO(A)    100.000000
 Name: count, dtype: float64),
'TIPO DA CONTA': (TIPO DA CONTA
 ESSENTIAL    666669
 VIP          261612
 PRIME        191719
 Name: count, dtype: int64,
 TIPO DA CONTA
 ESSENTIAL     59.524018
 VIP           82.882232
 PRIME        100.000000
 Name: count, dtype: float64),
'DEPENDENTES': (DEPENDENTES
 1    593164
 0    303341
 2    111989
 3     78375
 4     33131
 Name: count, dtype: int64,
 DEPENDENTES
 1     52.961071
 0     80.045089
 2     90.044107
 3     97.041875
 4    100.000000
 Name: count, dtype: float64),
'OPINIÃO DO CLIENTE': (OPINIÃO DO CLIENTE
 Entrega foi boa e rápida na UVVine                                  1
72818
 Funcionou bem e me atendeu no prazo certo a UVVine                  1
```

```
72639
  Fiquei satisfeito com o serviço da UVVine                                    1
72523
  Serviço eficiente, recomendo o clube:  UVVine                                1
72238
  Gostei do atendimento prestado pela UVVine                                   1
72182
  Nada digno de nota com a UVVine
33859
  Foi indiferente minha experiência com a UVVine
33673
  Não tenho muito a dizer sobre a UVVine
33549
  Sem opinião formada sobre a UVVine
33483
  Compro exponenciamente, opinião neutra sobre UVVine
33436
  Atendimento comum, poderia melhorar em cordialidade a UVVine
9056
  Nada demais, serviço regular na UVVine
9033
  Experiência regular, não atendeu a política de cashback da UVVine
9018
  Foi aceitável, esperava mais da UVVine
8857
  Razoável, com margem de melhora no preço médio da UVVine
8836
  Serviço impecável, adorei o clube:  UVVine
6825
  Superou expectativas o atendimento na UVVine
6762
  Ótimo! Tudo ocorreu maravilhosamente bem com a  UVVine
6721
  Produto perfeito, chegou antes do prazo de previsão. Top demais a UVVine
6657
  Excelente experiência, recomendo muito a UVVine
6635
  Senti que faltou qualidade no atendimento da UVVine
2287
  Tive uma experiência ruim com a UVVine
2258
  Não gostei do atendimento na UVVine
2251
  Serviço deixou a desejar na UVVine
2225
  Produto chegou atrasado na UVVine
2179
  Name: count, dtype: int64,
  OPINIÃO DO CLIENTE
  Entrega foi boa e rápida na UVVine
15.430179
  Funcionou bem e me atendeu no prazo certo a UVVine
30.844375
  Fiquei satisfeito com o serviço da UVVine
46.248214
  Serviço eficiente, recomendo o clube:  UVVine
61.626607
  Gostei do atendimento prestado pela UVVine
77.000000
  Nada digno de nota com a UVVine
```

```
80.023125
  Foi indiferente minha experiência com a UVVine
83.029643
  Não tenho muito a dizer sobre a UVVine
86.025089
  Sem opinião formada sobre a UVVine
89.014643
  Compro exponenciamente, opinião neutra sobre UVVine
92.000000
  Atendimento comum, poderia melhorar em cordialidade a UVVine
92.808571
  Nada demais, serviço regular na UVVine
93.615089
  Experiência regular, não atendeu a política de cashback da UVVine
94.420268
  Foi aceitável, esperava mais da UVVine
95.211071
  Razoável, com margem de melhora no preço médio da UVVine
96.000000
  Serviço impecável, adorei o clube:  UVVine
96.609375
  Superou expectativas o atendimento na UVVine
97.213125
  Ótimo! Tudo ocorreu maravilhosamente bem com a  UVVine
97.813214
  Produto perfeito, chegou antes do prazo de previsão. Top demais a UVVine
98.407589
  Excelente experiência, recomendo muito a UVVine
99.000000
  Senti que faltou qualidade no atendimento da UVVine
99.204196
  Tive uma experiência ruim com a UVVine
99.405804
  Não gostei do atendimento na UVVine
99.606786
  Serviço deixou a desejar na UVVine
99.805446
  Produto chegou atrasado na UVVine                                    1
00.000000
  Name: count, dtype: float64)}
```

Atividade A3: Resultado da Pesquisa de Satisfação

Fazer o resultado da Pesquisa de satisfação do cliente em porcentagem (%):

- Geral
- Por Sexo

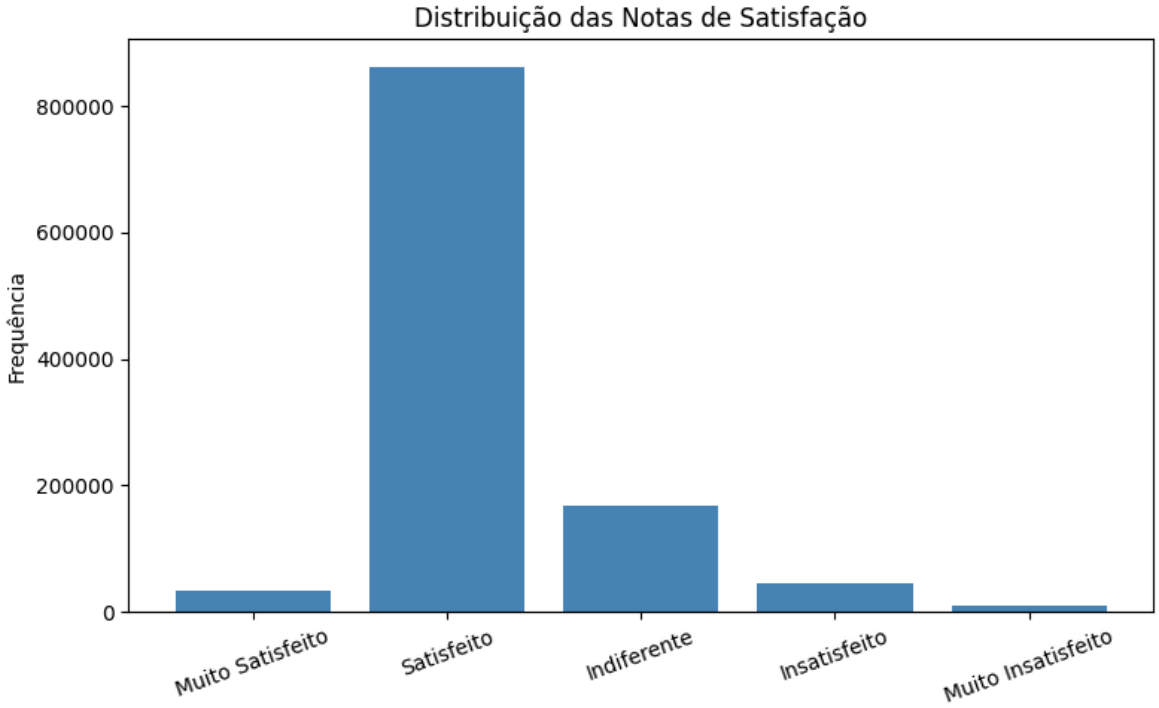


Atividade A3.1: Discretizar a Variável Qualitativa (OPINIÃO DO CLIENTE)

In [ ]:
```
AtividadeA3_1()
```

```
Valores não mapeados (deveria ser 0): 0

NOTA 5 (Muito Satisfeito): 33,600 (3.0%)
NOTA 4 (Satisfeito): 862,400 (77.0%)
NOTA 3 (Indiferente): 168,000 (15.0%)
NOTA 2 (Insatisfeito): 44,800 (4.0%)
NOTA 1 (Muito Insatisfeito): 11,200 (1.0%)
```



Out[ ]:

	NOTA
0	4
1	4
2	4
3	3
4	4
...	...
1119995	4
1119996	4
1119997	4
1119998	4
1119999	3

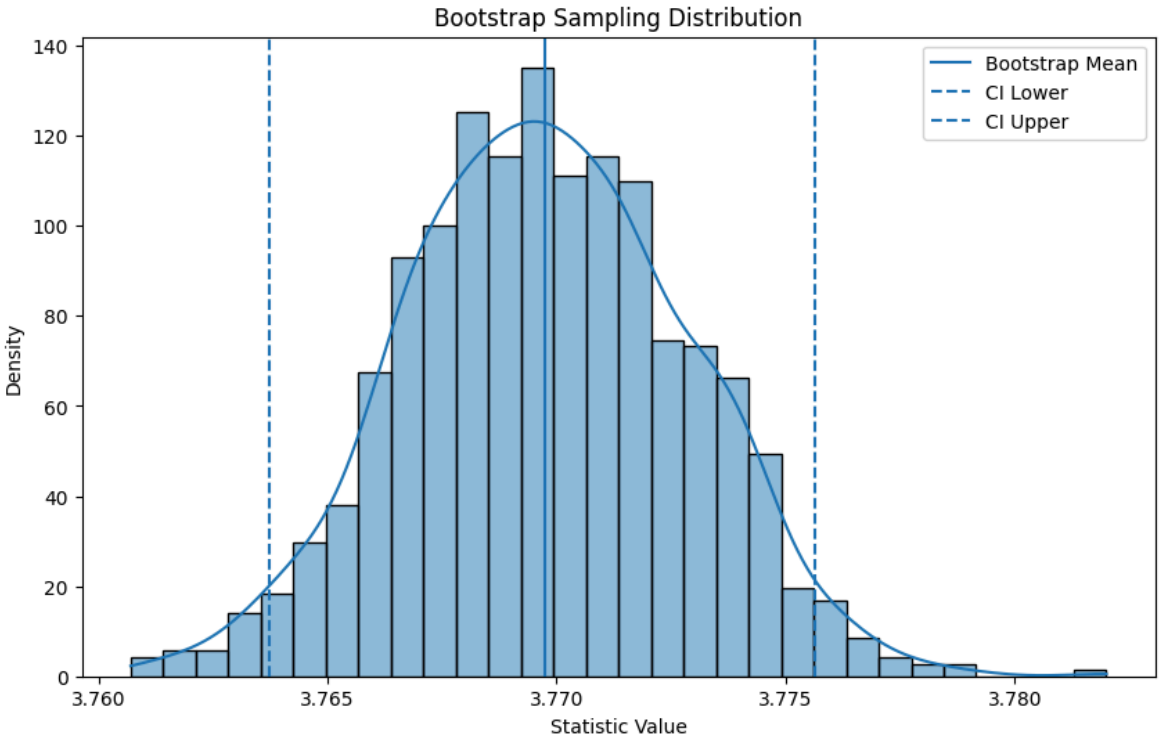
1120000 rows × 1 columns

dtype: int64

Atividade A3.2: Análise Bootstrap do resultado da Pesquisa de Satisfação

In [ ]:
```
AtividadeA3_2()
```

```
Média amostral da NOTA: 3.7696
```



```
CONFIDENCE INTERVAL
LOW  : 3.7637
HIGH : 3.7757
SE   : 0.0031

Erro-Padrão (bootstrap): 0.0031
IC 95%: [3.7637 ; 3.7757]
```

Out[ ]:
```
(605510     4
 969981     4
 912275     4
 1061009    4
 43855      4
            ..
 143678     1
 946988     4
 177062     3
 1013525    3
 995174     2
 Name: NOTA, Length: 40000, dtype: int64,
 BootstrapResult(confidence_interval=ConfidenceInterval(low=np.float64(3.763723
75), high=np.float64(3.77565)), bootstrap_distribution=array([3.774825, 3.7671
, 3.772125, 3.7682  , 3.7647  , 3.769975,
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        3.768925, 3.773575, 3.7674  , 3.771825, 3.7644  , 3.774525,
        3.770225, 3.770875, 3.765825, 3.7695  , 3.7669  , 3.76625 ,
        3.771025, 3.76705 , 3.768775, 3.76695 , 3.771825, 3.774375,
        3.76435 , 3.774075, 3.77355 , 3.771125, 3.764425, 3.77845 ,
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        3.77035 , 3.771925, 3.76885 , 3.773225, 3.771225, 3.768375,
        3.773375, 3.775325, 3.769525, 3.768575, 3.77065 , 3.7745  ,
        3.775575, 3.76905 , 3.7647  , 3.768075, 3.766225, 3.766875,
        3.76645 , 3.77165 , 3.774375, 3.7706  , 3.769625, 3.773175,
        3.773225, 3.774175, 3.768825, 3.768   , 3.7686  , 3.76705 ,
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        3.770725, 3.769875, 3.7728  , 3.77225 , 3.7708  , 3.765025,
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        3.767925, 3.767475, 3.772   , 3.7654  , 3.77065 , 3.7726  ,
        3.76995 , 3.7746  , 3.772675, 3.774925, 3.77205 , 3.776625,
        3.7733  , 3.77035 , 3.77115 , 3.771775, 3.768375, 3.77105 ,
        3.7655  , 3.774425, 3.76765 , 3.770275, 3.76825 , 3.7684  ,
        3.768   , 3.76985 , 3.76665 , 3.768775, 3.771475, 3.771375,
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        3.775775, 3.77435 , 3.77805 , 3.772275, 3.769   , 3.7678  ,
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3.765175, 3.7675  , 3.76795 , 3.771   , 3.7735  , 3.7693  ,
```

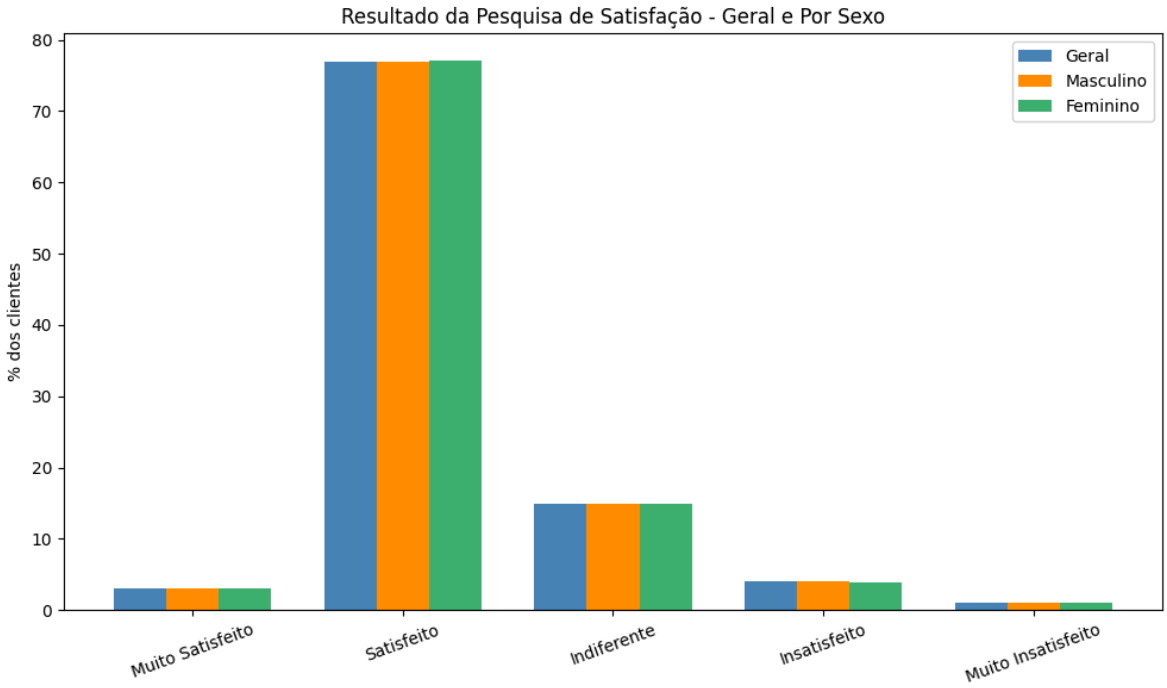
```
        3.76815 , 3.7668  , 3.7699  , 3.7718  ]), standard_error=np.float64(0.0
03064283615864152)))
```

Atividade A3.3: Resultado Final da Pesquisa de Satisfação

In [ ]:
```
AtividadeA3_3()
```

```
RESULTADO GERAL (%):
NOTA
5     3.0
4    77.0
3    15.0
2     4.0
1     1.0
Name: proportion, dtype: float64

RESULTADO POR SEXO (%):
NOTA          5      4      3     2     1
SEXO
FEMININO   3.01  77.03  14.99  3.95  1.02
MASCULINO  3.00  76.99  15.00  4.01  1.00
```



Out[ ]:
```
(NOTA
 5     3.0
 4    77.0
 3    15.0
 2     4.0
 1     1.0
 Name: proportion, dtype: float64,
 NOTA              5          4          3         2         1
 SEXO
 FEMININO   3.005864  77.033913  14.990913  3.948839  1.020471
 MASCULINO  2.999122  76.994924  15.001360  4.007658  0.996936)
```

Conclusão:

77% dos clientes estão satisfeitos, 3% muito satisfeitos, 15% indiferentes, e apenas 5% insatisfeitos ou muito insatisfeitos (somados). O resultado é praticamente idêntico entre

homens e mulheres (diferenças menores que 1 ponto percentual em cada nota) — o sexo do cliente não influencia a satisfação nessa base. A nota média de satisfação, validada por bootstrap, é de 3,77 (IC 95%: 3,76–3,78), numa escala de 1 a 5.

RELATÓRIO FINAL - PARTE 1

RELATÓRIO FINAL — PARTE 1

Resposta à pergunta central: Qual é o Tamanho Ideal da Amostra?

Testamos dois procedimentos pedidos no roteiro (empírico e teórico-empírico), que divergiram fortemente entre si — o teórico-empírico (Neyman, com threshold percentual) convergiu sempre perto do mesmo valor (~5.300), independentemente da variabilidade real dos dados, uma limitação matemática do próprio critério. Por isso, adotamos uma validação mais rigorosa e defensável: uma busca por bootstrap real, que encontrou o menor tamanho de amostra que garante margem de erro do Intervalo de Confiança de 95% ≤ 1% em cada grupo (**VIP: 10 mil; PRIME: 15 mil; ESSENTIAL: 25 mil; GERAL, sem separar por grupo: 40 mil**).

Adotamos **n = 40.000** por grupo — o cenário mais exigente — validado por três propriedades estatísticas independentes: **viés praticamente nulo** (< 0,02% em todos os grupos), **margem de erro abaixo de 1%**, e **cobertura real do Intervalo de Confiança entre 95% e 98%** (checada por simulação direta contra a média verdadeira da população, algo possível aqui porque temos acesso à população inteira, não só a uma amostra).

Síntese da Parte 1:

- A base (1,12 milhão de notas fiscais) foi tratada (1 linha corrompida removida) e está pronta para uso.
- O tamanho de amostra de 40.000 por grupo é estatisticamente sólido e será a referência de trabalho daqui pra frente.
- ESSENTIAL e PRIME são, comprovadamente, perfis de consumo distintos em todos os aspectos testados (distribuição, média, variabilidade).
- Os dados não seguem distribuição normal — relevante para a escolha de modelos no Workflow 2 (Machine Learning).
- Há concentração estrutural forte em quase todas as variáveis qualitativas (região, sexo, tipo de conta) — classes desbalanceadas que podem exigir tratamento específico na modelagem.
- A satisfação do cliente é majoritariamente positiva (80% entre satisfeito e muito satisfeito) e não varia por sexo.
- O perfil químico do vinho contém ruído de simulação (valores impossíveis) — uma limitação conhecida e documentada da base, não um erro de análise.

In [ ]:
```
!jupyter nbconvert --to html "/content/MarcioJr_PARTE_1.ipynb"
```

```
[NbConvertApp] Converting notebook /content/MarcioJr_PARTE_1.ipynb to html
[NbConvertApp] WARNING | Alternative text is missing on 23 image(s).
[NbConvertApp] Writing 4268831 bytes to /content/MarcioJr_PARTE_1.html
```